



L_1 Geodesic Farthest Neighbors in a Simple Polygon and Related Problems

Sang Won Bae¹ 

Received: 16 September 2017 / Revised: 19 May 2019 / Accepted: 6 June 2019 / Published online: 17 June 2019
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Abstract

We investigate the L_1 geodesic farthest neighbors in a simple polygon P , and address several fundamental problems related to farthest neighbors. Given a subset $S \subseteq P$, an L_1 geodesic farthest neighbor of $p \in P$ from S is one that maximizes the length of L_1 shortest path from p in P . Our list of problems include: computing the diameter, radius, center, farthest-neighbor Voronoi diagram, and two-center of S under the L_1 geodesic distance. We show that all these problems can be solved in linear or near-linear time based on our new observations on farthest neighbors and extreme points. Among them, the key observation shows that there are at most four extreme points of any compact subset $S \subseteq P$ with respect to the L_1 geodesic distance after removing redundancy.

Keywords L_1 geodesic distance · Simple polygon · Farthest neighbor · Diameter · Radius · Center · Two-center

Mathematics Subject Classification 68U05 · 68Q25 · 68W05

1 Introduction

The geometry of points in a simple polygon P has been one of the most attractive research subjects in computational geometry since the 1980s. As a metric space, P is often associated with a distance function d induced by shortest paths that stay inside

Editor in Charge: Kenneth Clarkson

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2015R1D1A1A01057220 and 2018R1D1A1B07042755). A preliminary version of this work was presented at ISAAC 2016.

Sang Won Bae
swbae@kgu.ac.kr

¹ Division of Computer Science and Engineering, Kyonggi University, Suwon, Korea

P . Indeed, there are several ways to define a shortest path between two points in P , depending on which underlying metric is adopted to determine the length of a segment in P . Most common are the Euclidean and the L_1 metrics that define the Euclidean and the L_1 shortest paths, respectively, in P . The length of a shortest path between two points $p, q \in P$ is called the (Euclidean or L_1) geodesic distance $d(p, q)$.

In this paper, we are interested in fundamental problems related to geodesic farthest neighbors in P . Given a set S of points in P , a farthest neighbor of $p \in P$ from S is one that maximizes the geodesic distance $d(p, q)$ from p to every $q \in S$. Specifically, our list of problems includes those of computing the following:

- The *farthest-neighbor Voronoi diagram* of S .
- The *diameter* of S : $\text{diam}(S) := \max_{q \in S} \max_{q' \in S} d(q, q')$.
- The *radius* of S : $\text{rad}(S) := \min_{p \in P} \max_{q \in S} d(p, q)$.
- A *center* of S : a point $c \in P$ such that $\max_{q \in S} d(c, q) = \text{rad}(S)$.
- A *two-center* of S : a pair of points $c_1, c_2 \in P$ that minimizes $\max_{q \in S} \min\{d(c_1, q), d(c_2, q)\}$.

In the Euclidean case, where d denotes the Euclidean geodesic distance, these problems have been intensively studied. Aronov et al. [2] presented an $O((n+m) \log(n+m))$ -time algorithm that computes the farthest-neighbor Voronoi diagram of S if S consists of m points and P is an n -gon. Recently, Oh et al. [12] showed that the diagram can be computed faster in $O(n \log \log n + m \log(n+m))$ time, or in $O(n \log \log n)$ time when S is the set of vertices of P . After the preparation of this paper, an optimal-time randomized algorithm is now known by Barba [4]: in $O(n+m \log m)$ time, or in $O(n)$ time when S is a subset of the vertices of P . Note that computing the diameter, radius, and center of S is reduced from the farthest-neighbor Voronoi diagram in linear time. In a special case where $S = P$, it is known that we can compute them in linear $O(n)$ time without computing the Voronoi diagram [1, 10]. The problem of computing a two-center of S under the Euclidean geodesic distance was recently addressed by Oh et al. [14] and Oh et al. [13], resulting in two algorithms that run in $O(n^2 \log^2 n)$ time when $S = P$ and in $O(m(m+n) \log^3(m+n))$ time when S is a set of m points in P .

The problems in the case of L_1 geodesic distance have attained less interest compared to those in the Euclidean case. This is probably because most of results for the Euclidean counterpart automatically hold for the L_1 geodesic distance. Note that the Euclidean shortest paths in P are also L_1 shortest paths, and the algorithm of Aronov et al. [2] can be implemented for computing the L_1 geodesic farthest-neighbor Voronoi diagram. However, it is not clear whether the approach by Oh et al. [12] or by Barba [4] can be extended to compute the L_1 diagram. Bae et al. [3] exhibited some geometric observations on the L_1 geodesic distance that are different from the Euclidean one, and exploited them to devise linear-time algorithms that compute the diameter, radius, and center of a simple polygon P , i.e., the special case where $S = P$. Prior to this work, no algorithm for the two-center of S in the L_1 geodesic distance was known in the literature.

In this paper, we reveal that farthest neighbors in the L_1 geodesic distance behave quite different from—indeed much nicer than—those in the Euclidean geodesic dis-

tance. Based on our new observations, we show that all the problems listed above in the L_1 geodesic distance can be computed in linear or near-linear time:

- $O(n + m \log n)$ time when S is a set of m points in P , or
- $O(n)$ time when either $S = P$ or S equals the set of vertices of P .

It is worth noting that our algorithms run in time linear to each of n and m , while the $O(m \log n)$ term is due to $O(m)$ evaluations of the L_1 geodesic distance $d(p, q)$, which was unavoidable in our algorithm. Note that, in particular, our algorithm for the two-center is faster than the currently best algorithm in the Euclidean case, roughly by a factor of n or m [13,14].

All these algorithmic results are based on our new discoveries on L_1 geodesic farthest neighbors. Some remarkable observations compared to the Euclidean counterpart are as follows:

- There are at most four extreme points of S ; while there can be linearly many extreme points in the Euclidean setting.
- The L_1 geodesic diameter of S is exactly twice its radius; while this is not always true in the Euclidean setting.
- The set of all centers of S forms a line segment in P ; while there is a unique center of S in the Euclidean setting.
- The farthest-neighbor Voronoi diagram is of $O(1)$ complexity; while it has linear complexity in the Euclidean setting.

Note that we prove the above properties in a general setting for any arbitrary compact subset S of P . Consequently, the L_1 geodesic diameter, radius, center, farthest-neighbor Voronoi diagram, and even a two-center of any compact subset S can be computed in $O(n)$ time, provided the four extreme points of S .

This phenomenon can be understood as an extension of the relation between the L_1 and the Euclidean distances in the whole plane $\mathbb{R}^2$. In the plane associated with the L_1 metric, there are at most four extreme points of S in the four directions corresponding to the four segments of the L_1 metric balls, while in the plane associated with the Euclidean metric, every point of S lying on the boundary of its convex hull is considered to be extreme. An immediate implication is that the farthest-neighbor Voronoi diagram in the L_1 metric consists of at most four nonempty regions and thus has $O(1)$ complexity. This further implies that the farthest-neighbor Voronoi diagram, diameter, radius, and center of S under the L_1 metric can be computed in $O(1)$ time after the four extreme points of S are obtained in $O(m)$ time. For the plane with the Euclidean metric, computing the extreme points, the farthest-neighbor Voronoi diagram, and even the diameter of S takes $O(m \log m)$ time and has a lower bound of $\Omega(m \log m)$ [15], while the radius and the center of S can be computed in $O(m)$ time [11]. Similarly, an L_1 (or rectilinear) two-center of m points in the plane can be computed in $O(m)$ time [6], while the best known algorithm that computes a Euclidean two-center in the plane runs in $O(m \log^2 m (\log \log m)^2)$ deterministic time [5]. Our results thus provide a series of analogies on farthest neighbors in the plane with the L_1 metric into those in the metric space (P, d) , where P is a simple polygon and d is the L_1 geodesic distance in P .

The rest of the paper is organized as follows. After some preliminaries are given in Sect. 2, several useful properties of L_1 shortest paths are discussed in Sect. 3.

Section 4 is devoted to investigating L_1 geodesic farthest neighbors and extreme points, and shows that there are at most four extreme points in any compact subset of a given simple polygon under the L_1 geodesic distance. Next, the L_1 geodesic center and farthest-neighbor Voronoi diagram are discussed in Sects. 5 and 6, respectively. In Sect. 7, we present our algorithms that compute the extreme points, the diameter, the radius, the center, and the farthest-neighbor Voronoi diagram for a given compact subset. Finally, we solve the L_1 geodesic two-center problem in Sect. 8.

2 Preliminaries

For any subset $A \subset \mathbb{R}^2$, we denote by ∂A and $\text{int } A$ the boundary and the interior of A , respectively. For $p, q \in \mathbb{R}^2$, denote by pq the line segment with endpoints p and q . For any path π in $\mathbb{R}^2$, let $|\pi|$ be the length of π under the L_1 metric¹, or simply the L_1 length. Note that $|pq|$ equals the L_1 distance between p and q .

The following is a basic observation on the L_1 length of paths in $\mathbb{R}^2$. A path is called *XY-monotone*, or simply *monotone* throughout the paper, if any vertical or horizontal line intersects it in at most one connected component.

Lemma 2.1 *Let $p, q \in \mathbb{R}^2$ be any two points and π be any path between p and q . Then, $|\pi| = |pq|$ if and only if π is monotone.*

Proof Let x_p and y_p be the x - and y -coordinates of p , respectively, and x_q and y_q be the x - and y -coordinates of q , respectively. Without loss of generality, assume that $x_p \leq x_q$ and $y_p \leq y_q$.

First, we consider the case where π is polygonal, so π consists of k vertices $p = v_1, v_2, \dots, v_k = q$ for some $k \geq 2$. Let x_i and y_i be the x - and y -coordinates of v_i for $i = 1, \dots, k$. Then, the L_1 length $|\pi|$ of π is equal to

$$|\pi| = \sum_{i=1, \dots, k-1} |v_i v_{i+1}| = \sum_{i=1, \dots, k-1} (|x_{i+1} - x_i| + |y_{i+1} - y_i|).$$

Hence, $|\pi| = |pq| = (x_q - x_p) + (y_q - y_p) = (x_k - x_1) + (y_k - y_1)$ if and only if it holds that $x_1 \leq x_2 \leq \dots \leq x_k$ and $y_1 \leq y_2 \leq \dots \leq y_k$, which is equivalent to the condition that π is monotone.

Next, we consider the general case where π may not be polygonal. Consider any sequence of polygonal paths $\pi_1, \pi_2, \dots$ between p and q that converges to π . Note that $|\pi| = \lim_{i \rightarrow \infty} |\pi_i|$.

Suppose that π is monotone. Then, there exists a subsequence $\pi'_1, \pi'_2, \dots$ of $\pi_1, \pi_2, \dots$ such that $\lim_{i \rightarrow \infty} \pi'_i = \pi$ and each π'_i is polygonal and monotone. Since each π'_i is polygonal and monotone, we have $|\pi'_i| = |pq|$ on the one hand. On the other hand, we have $|\pi| = \lim_{i \rightarrow \infty} |\pi_i| = \lim_{i \rightarrow \infty} |\pi'_i|$. This concludes that $|\pi| = |pq|$.

If π is not monotone, then here exists a subsequence $\pi'_1, \pi'_2, \dots$ of $\pi_1, \pi_2, \dots$ such that $\lim_{i \rightarrow \infty} \pi'_i = \pi$ and each π'_i is not monotone. Since each π'_i is not monotone,

¹ If π is polygonal, then $|\pi|$ is the sum of the L_1 distances between its consecutive vertices. In general, $|\pi|$ is the limit of the sequence $|\pi_1|, |\pi_2|, \dots$ such that each π_i for $i = 1, 2, \dots$ is a polygonal path and the sequence $\pi_1, \pi_2, \dots$ converges to π .

we have $|\pi'_i| > |pq|$. This implies that $|\pi| = \lim_{i \rightarrow \infty} |\pi_i| = \lim_{i \rightarrow \infty} |\pi'_i| > |pq|$, concluding the lemma. $\square$

Let P be a simple polygon with n vertices. We regard P as a compact subset in $\mathbb{R}^2$, so its boundary ∂P is contained in P . An L_1 shortest path between p and q is a path joining p and q that lies in P and minimizes its L_1 length. The L_1 geodesic distance $d(p, q)$ is the L_1 length of an L_1 shortest path between p and q . For any $p, q \in P$, let $\Pi(p, q)$ be the set of all L_1 shortest paths from p to q .

Analogously, a path lying in P minimizing its Euclidean length is called the *Euclidean shortest path*. It is well known that there is always a unique Euclidean shortest path between any two points in a simple polygon [8]. We let $\pi_2(p, q)$ be the unique Euclidean shortest path from $p \in P$ to $q \in P$. The following states a crucial relation between Euclidean and L_1 shortest paths in a simple polygon.

Lemma 2.2 (Hershberger and Snoeyink [9]) *For any two points $p, q \in P$, the Euclidean shortest path $\pi_2(p, q)$ is also an L_1 shortest path between p and q . That is, $\pi_2(p, q) \in \Pi(p, q)$.*

Lemma 2.2 enables us to exploit several structures for Euclidean shortest paths such as Guibas et al. [8] and Guibas and Hershberger [7].

Another important concept regarding the shortest paths in P is the relative convexity. A subset $A \subset P$ is called *relatively convex* if $\pi_2(p, q) \subset A$ for any $p, q \in A$. For any subset $A \subset P$, the *relative convex hull* $\text{rconv}(A)$ of A is the smallest relatively convex set that contains A . If A is the set of m points in P , then its relative convex hull forms a weakly simple polygon in P with $O(m + n)$ vertices. Touissant [16] presented an $O((n + m) \log(n + m))$ -time algorithm that computes $\text{rconv}(A)$, and Guibas and Hershberger [7] improved it to $O(n + m \log(n + m))$.

Throughout the paper, unless otherwise stated, a shortest path and the geodesic distance always refer to an L_1 shortest path and the L_1 geodesic distance d .

3 Properties of L_1 Shortest Paths

In this section, we observe several useful properties of L_1 shortest paths in P .

We define a *chord* of P to be a maximal segment contained in P . For any $z \in P$, let h_z^- and h_z^+ be the left and right endpoints, respectively, of the horizontal chord through z , while v_z^- and v_z^+ denote the lower and upper endpoints, respectively, of the vertical chord through z . Note that the horizontal or vertical chord may intersect the boundary ∂P of P in several connected components by definition. We also consider the four segments zh_z^- , zh_z^+ , zv_z^- and zv_z^+ , called the *leftward*, *rightward*, *downward*, and *upward half-chords* from z , respectively.

Let $z \in P$ be fixed and $p \in P$ be any point. We say that $\pi \in \Pi(p, z)$ *chooses* a half-chord from z if π intersects it at a point other than z . Then, Lemma 2.1 implies that every $\pi \in \Pi(p, z)$ chooses at most one half-chord from z or none of the four. We then observe the following.

Lemma 3.1 *For any $p, z \in P$, there are no two shortest paths $\pi, \pi' \in \Pi(p, z)$ such that π chooses a half-chord from z and π' chooses its opposite half-chord from z .*

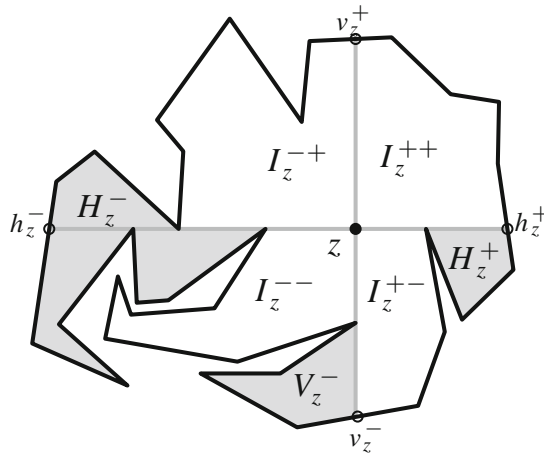


Fig. 1 Partition around z

Proof We consider the partition of P into P^- and P^+ by the horizontal chord $h_z^- h_z^+$ through z , where $P^- \subset P$ is the set of points that are bounded by $h_z^- h_z^+$ and the chain along ∂P from h_z^- to h_z^+ in the counter-clockwise direction and $P^+ \subset P$ is the set of points that are bounded by $h_z^- h_z^+$ and the chain along ∂P from h_z^+ to h_z^- in the counter-clockwise direction.

Assume there is $\pi \in \Pi(p, z)$ that chooses the downward half-chord $z v_z^-$. By definition, this means that $\pi \cap z v_z^-$ contains a point other than z . This implies by Lemma 2.1 that p must lie in P^- and $p \notin h_z^- h_z^+$.

Suppose that there is another shortest path $\pi' \in \Pi(p, z)$ that chooses the upward half-chord $z v_z^+$. Then, by the same argument, p must lie in P^+ , a contradiction. $\square$

This implies that any $p \in P$ avoids at least two half-chords that are not opposite by shortest paths to z . We thus consider the four regions of P according to the pair of excluded half-chords. More precisely, for any $\sigma_1, \sigma_2 \in \{+, -\}$, let $P_z^{\sigma_1 \sigma_2} \subset P$ be the set of points $p \in P$ such that no shortest path $\pi \in \Pi(p, z)$ chooses $z h_z^{\sigma_1}$ or $z v_z^{\sigma_2}$, where $\bar{+} = -$ and $\bar{-} = +$. Lemma 3.1 guarantees that $P = P_z^{--} \cup P_z^{-+} \cup P_z^{+-} \cup P_z^{++}$ for any $z \in P$, while these four regions are not disjoint. Also, note that $P_z^{--} \cap P_z^{++} = \{z\}$ and $P_z^{-+} \cap P_z^{+-} = \{z\}$.

In order to gain a comprehensive understanding on the four regions $P_z^{\sigma_1 \sigma_2}$, we consider the following eight subsets of P around z :

$$\begin{aligned} H_z^{\sigma_1} &:= P_z^{\sigma_1 -} \cap P_z^{\sigma_1 +}, \\ V_z^{\sigma_2} &:= P_z^{-\sigma_2} \cap P_z^{+\sigma_2}, \text{ and} \\ I_z^{\sigma_1 \sigma_2} &:= P_z^{\sigma_1 \sigma_2} \setminus (H_z^{\sigma_1} \cup V_z^{\sigma_2}), \end{aligned}$$

for each $\sigma_1, \sigma_2 \in \{+, -\}$. Note that $P_z^{\sigma_1 \sigma_2} = I_z^{\sigma_1 \sigma_2} \cup H_z^{\sigma_1} \cup V_z^{\sigma_2}$. Observe that H_z^- , for example, is the set of points $p \in P$ such that no shortest path in $\Pi(p, z)$ chooses the downward, upward, or rightward half-chord from z , and I_z^{--} is the set of points $p \in P$

that admit two shortest paths $\pi, \pi' \in \Pi(p, z)$ such that π chooses the leftward half-chord and π' chooses the downward half-chord from z . See Fig. 1 for an illustration. Note that these eight subsets $H_z^{\sigma_1}$, $V_z^{\sigma_2}$, and $I_z^{\sigma_1\sigma_2}$ form a partition of $P \setminus \{z\}$ around z . In most cases where the horizontal and vertical chords through z intersect ∂P only at their endpoints, we have $H_z^{\sigma_1} = zh_z^{\sigma_1}$ and $V_z^{\sigma_2} = zv_z^{\sigma_2}$. However, this is not always the case, as shown in Fig. 1, specifically, where H_z^-, H_z^+ and V_z^- include the shaded regions, plus their corresponding half-chords from z .

To be more precise, consider the complement $\mathcal{C}_z := P \setminus (h_z^- h_z^+ \cup v_z^- v_z^+)$, which in general consists of several connected components. Such a component $C \subseteq \mathcal{C}_z$ is said to be *adjacent* to a half-chord from z if its boundary ∂C intersects the half-chord at a point other than z . Note that any component of $\mathcal{C}_z$ is adjacent to at least one and at most two half-chords from z . The following describes how $H_z^-, H_z^+, V_z^-,$ and V_z^+ are formed.

Lemma 3.2 *Let $z \in P$ and $\sigma_1, \sigma_2 \in \{+, -\}$. Then, $H_z^{\sigma_1}$ is equal to the union of $zh_z^{\sigma_1}$ and the components of $\mathcal{C}_z$ that are adjacent to $zh_z^{\sigma_1}$ but to none of $H_z^{\sigma_1}, V_z^+,$ and V_z^- . Analogously, $V_z^{\sigma_2}$ is equal to the union of $zv_z^{\sigma_2}$ and the components of $\mathcal{C}_z$ that are adjacent to $zv_z^{\sigma_2}$ but to none of $H_z^+, H_z^-,$ and $V_z^{\sigma_2}$.*

Proof Consider any component C of $\mathcal{C}_z = P \setminus (h_z^- h_z^+ \cup v_z^- v_z^+)$. Note that C is adjacent to one or two half-chords from z .

We consider the leftward half-chord zh_z^- and the corresponding set H_z^- . The other three can be handled in a symmetrical way. If C is not adjacent to the leftward half-chord zh_z^- , then for any $p \in C$, no shortest path $\pi \in \Pi(p, z)$ chooses the leftward half-chord zh_z^- , so $C \cap H_z^- = \emptyset$, in this case.

If C is adjacent to zh_z^- but to none of the other three half-chords, then every shortest path $\pi \in \Pi(p, z)$ for any $p \in C$ cannot choose the other three half-chords from z , namely, the upward, downward, and rightward half-chords. This implies that any $p \in C$ belongs to P_z^{--} and simultaneously to P_z^{-+} by definition. Since $H_z^- = P_z^{--} \cap P_z^{-+}$, this implies that $C \subseteq H_z^-$.

Now, suppose that C is adjacent to zh_z^- and also to another half-chord from z , say the downward half-chord zv_z^- . Then, we have $z \in \partial C$, and $v_z^- \neq z$. Pick any point $p \in C$ and consider the Euclidean shortest path $\pi = \pi_2(p, z)$. Note by Lemma 2.2 that $\pi \in \Pi(p, z)$. Let w be the last vertex of P through π from p to z , that is, $\pi = \pi_2(p, w) \cup wz$. By Lemma 2.1, we have $w \in C$. Then, there exists a point $w' \in zv_z^-$ with $w' \neq z$ such that $\pi_2(w, w')$ is monotone. See Fig. 2. Now, we consider another path π' from p to z obtained from π by replacing the last segment wz by $\pi_2(w, w') \cup w'z$. Since the substitute $\pi_2(w, w') \cup w'z$ is monotone, it is an L_1 shortest path from w to z by Lemma 2.1. So, $\pi' \in \Pi(p, z)$ is another shortest path from p to z . Moreover, π' chooses the downward half-chord zv_z^- from z since $w' \in zv_z^-$ and $w' \neq z$. This implies that $p \neq H_z^-$, and thus that $C \cap H_z^- = \emptyset$, in this case. This completes the proof of the lemma. $\square$

Thus, any component C of $\mathcal{C}_z$ that is adjacent to exactly one half-chord is included into the corresponding subset $H_z^{\sigma_1}$ or $V_z^{\sigma_2}$ for some $\sigma_1, \sigma_2 \in \{+, -\}$. On the other hand, if a component C of $\mathcal{C}_z$ is adjacent to two half-chords from z , then the boundary of C must contain z . Thus, there are at most four such components of $\mathcal{C}_z$, and each

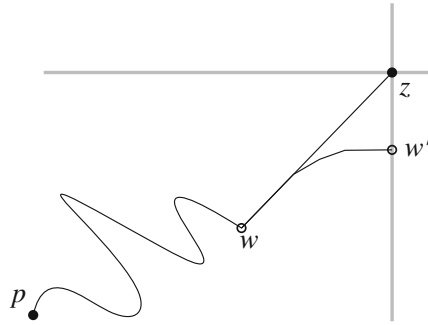


Fig. 2 Illustration to the proof of Lemma 3.2

of them forms $I_z^{\sigma_1\sigma_2}$ for some $\sigma_1, \sigma_2 \in \{+, -\}$. Lemma 3.2 and the above discussion imply the following corollary.

Corollary 3.3 *Suppose that $H_z^{\sigma_1} \setminus zh_z^{\sigma_1} \neq \emptyset$ for $\sigma_1 \in \{+, -\}$. Then, either $z \in \{v_z^-, v_z^+\}$ or there exists a vertex u of P lying on zh_z^- such that for all $p \in H_z^{\sigma_1} \setminus zh_z^{\sigma_1}$ any shortest path $\pi \in \Pi(p, z)$ passes through u . An analogous claim also holds for the set $V_z^{\sigma_2}$ with $\sigma_2 \in \{+, -\}$.*

Proof We consider the case of H_z^- , as the other case is symmetric. Suppose to the contrary that $z \notin \{z_z^-, z_z^+\}$ and there no vertex of P on the leftward half-chord zh_z^- from z . Then, there are exactly two connected components of C_z that are adjacent to the leftward half-chord zh_z^- from z . Moreover, these two components are exactly I_z^{--} and I_z^{-+} as discussed above. This implies that there is no component of C_z that is adjacent to zh_z^- but to none of the other three half-chords, so we have $H_z^- = zh_z^-$ by Lemma 3.2.

Therefore, if $H_z^- \setminus zh_z^- \neq \emptyset$, then $z \in \{z_z^-, z_z^+\}$ or there is a vertex of P on the leftward half-chord zh_z^- from z . In the latter case, we let u be the vertex on zh_z^- that is the closest from z . Then, it is clear that for any $p \in H_z^- \setminus zh_z^-$, any shortest path from p to z contains the segment uz . $\square$

We then prove the following properties of L_1 shortest paths in terms of the partition around a point $z \in P$.

Lemma 3.4 *Let $z \in P$ and A be any one of the following regions: $H_z^{\sigma_1}$, $V_z^{\sigma_2}$, $I_z^{\sigma_1\sigma_2}$, $I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}$, $I_z^{\sigma_1\sigma_2} \cup H_z^{\sigma_1}$, $I_z^{\sigma_1\sigma_2} \cup V_z^{\sigma_2}$, and $P_z^{\sigma_1\sigma_2}$ for any $\sigma_1, \sigma_2 \in \{+, -\}$. Then, for any $p, q \in A$, every $\pi \in \Pi(p, q)$ is contained in A .*

Proof First, note that Lemma 2.1 implies that for any $p, q \in P$ and $\pi \in \Pi(p, q)$, π intersects any vertical or horizontal chord through $z \in P$ in a connected subset. In this proof, we consider the connected components of $C_z = P \setminus (h_z^- h_z^+ \cup v_z^- v_z^+)$.

For any two connected components C and C' of C_z , and for any $p \in C$ and $q \in C'$, we show the following properties.

1. If $C = C'$, then every $\pi \in \Pi(p, q)$ is contained in C .
2. If both C and C' are adjacent to half-chord $zh_z^{\sigma_1}$ from z , then every $\pi \in \Pi(p, q)$ is contained in $C \cup C' \cup zh_z^{\sigma_1}$.

If $C = C'$, then $p, q \in C$. Suppose to the contrary that π intersects $h_z^- h_z^+ \cup v_z^- v_z^+$. Let $u, w \in \pi$ be the first and last point in the intersection $\pi \cap (h_z^- h_z^+ \cup v_z^- v_z^+)$ along π from p to q . Then, we have $u, w \in \partial C \cap (h_z^- h_z^+ \cup v_z^- v_z^+)$ as C is a component of $\mathcal{C}_z$. Also, note that the subpath π_u of π from p to u and another subpath π_w of π from w to q are contained in C , except u and w . Pick a point $u' \in \pi_u$ that is arbitrarily close to u and another $w' \in \pi_w$ that is arbitrarily close to w . Then, observe that the subpath of π from u' to w' is not monotone, so π is not a shortest path by Lemma 2.1, a contradiction. This implies that no $\pi \in \Pi(p, q)$ intersects $h_z^- h_z^+ \cup v_z^- v_z^+$, and thus that every $\pi \in \Pi(p, q)$ is contained in C if $p, q \in C$.

Now, suppose that both C and C' are adjacent to $zh_z^{\sigma_1}$. If $C = C'$, then every $\pi \in \Pi(p, q)$ is contained in C by property 1, so we are done. Otherwise, if $C \neq C'$, then any $\pi \in \Pi(p, q)$ must intersect the half-chord $zh_z^{\sigma_1}$. On the other hand, as observed above, $\pi \cap zh_z^{\sigma_1}$ is a connected subset uw , where $u \in \partial C$ and $w \in \partial C'$. We know that the subpath of π from p to u , except u , is contained in C , and the subpath of π from w to q , except w , is contained in C' by property 1. On the other hand, we have $uw \subset zh_z^{\sigma_1}$. This implies that π is contained in $C \cup C' \cup zh_z^{\sigma_1}$. Analogously, if both C and C' are adjacent to $zv_z^{\sigma_2}$, then every $\pi \in \Pi(p, q)$ is contained in $C \cup C' \cup zv_z^{\sigma_2}$.

Now, we prove the lemma for A , being each of

$$H_z^{\sigma_1}, V_z^{\sigma_2}, I_z^{\sigma_1\sigma_2}, I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}, I_z^{\sigma_1\sigma_2} \cup H_z^{\sigma_1}, I_z^{\sigma_1\sigma_2} \cup V_z^{\sigma_2}, P_z^{\sigma_1\sigma_2},$$

for any $\sigma_1, \sigma_2 \in \{+, -\}$.

Suppose that $p, q \in H_z^{\sigma_1}$. If both $p, q \in zh_z^{\sigma_1}$, then we have $\Pi(p, q) = \{pq\}$, so we are done. If both p and q do not lie on $zh_z^{\sigma_1}$, then this case is shown by properties 1 and 2. The remaining case is when one of p and q , say q , lies in $zh_z^{\sigma_1}$. Then, any $\pi \in \Pi(p, q)$ must intersect $zh_z^{\sigma_1}$ in a connected component wq . Observe by property 1 that the subpath from p to w , except w , is contained in the connected component C of $\mathcal{C}_z$ that contains p . Since $p \in H_z^{\sigma_1}$, C is adjacent to $zh_z^{\sigma_1}$ and to none of the other three by Lemma 3.2. This implies that π is contained in $C \cup zh_z^{\sigma_1} \subseteq H_z^{\sigma_1}$. Hence, if $A = H_z^{\sigma_1}$, then the lemma follows. By the analogous argument, this also proves the case when $A = V_z^{\sigma_2}$.

The case of $A = I_z^{\sigma_1\sigma_2}$ is shown by property 1 since $I_z^{\sigma_1\sigma_2}$ is a connected component of $\mathcal{C}_z$. Consider the case of $A = I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}$, and suppose that $p, q \in I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}$. If both p and q lie in $I_z^{\sigma_1\sigma_2}$, then we are done. Thus, suppose that one of p and q lies on $\partial I_z^{\sigma_1\sigma_2} \cap (zh_z^{\sigma_1} \cup zv_z^{\sigma_2})$. Without loss of generality, we assume that $p \in \partial I_z^{\sigma_1\sigma_2} \cap zh_z^{\sigma_1}$. If $q \in \partial I_z^{\sigma_1\sigma_2} \cap zh_z^{\sigma_1}$, then $\Pi(p, q) = \{pq\} \subset \partial I_z^{\sigma_1\sigma_2}$, so we are done. If $q \in \partial I_z^{\sigma_1\sigma_2} \cap zv_z^{\sigma_2}$, then $d(p, q) = d(p, z) + d(z, q)$, so every shortest path $\pi \in \Pi(p, q)$ must be monotone by Lemma 2.1. Such a path is contained in $I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}$. The remaining case is when $q \in I_z^{\sigma_1\sigma_2}$. Then, any shortest path $\pi \in \Pi(p, q)$ intersects $zh_z^{\sigma_1}$ in a connected subset pu . Observe that the subpath of π from u to q , except u , is contained in $I_z^{\sigma_1\sigma_2}$ by property 1. Since $pu \subset \partial I_z^{\sigma_1\sigma_2}$, we conclude that π is contained in $I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}$. This proves the lemma for $A = I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}$.

Suppose that $p, q \in I_z^{\sigma_1\sigma_2} \cup H_z^{\sigma_1}$. The case of $p, q \in H_z^{\sigma_1}$ has been handled above, while the case of $p, q \in I_z^{\sigma_1\sigma_2}$ is handled by property 1, since $I_z^{\sigma_1\sigma_2}$ is a connected component of $\mathcal{C}_z$. Thus, we assume that $p \in I_z^{\sigma_1\sigma_2}$ and $q \in H_z^{\sigma_1}$. If $q \in zh_z^{\sigma_1}$, then any $\pi \in \Pi(p, q)$ intersects $zh_z^{\sigma_1}$ in a connected component uq . Then, we know

that the subpath of π from p to u , except u , is contained in $I_z^{\sigma_1\sigma_2}$. This implies that π is contained in $I_z^{\sigma_1\sigma_2} \cup H_z^{\sigma_1}$. Otherwise, if $q \notin zh_z^{\sigma_1}$, then q lies in a connected component C' that is adjacent to $zh_z^{\sigma_1}$ but to none of the others by Lemma 3.2. Since $I_z^{\sigma_1\sigma_2}$ is another connected component of $\mathcal{C}_z$ that is adjacent to $zh_z^{\sigma_1}$, this is the case of property 2. Thus, in either case, every $\pi \in \Pi(p, q)$ is contained in $I_z^{\sigma_1\sigma_2} \cup H_z^{\sigma_1}$. This proves the lemma for $A = I_z^{\sigma_1\sigma_2} \cup H_z^{\sigma_1}$, and also for $A = I_z^{\sigma_1\sigma_2} \cup V_z^{\sigma_2}$ by symmetry.

Note that $P_z^{\sigma_1\sigma_2} = I_z^{\sigma_1\sigma_2} \cup H_z^{\sigma_1} \cup V_z^{\sigma_2}$. When both p and q lie in either $I_z^{\sigma_1\sigma_2} \cup H_z^{\sigma_1}$, or $I_z^{\sigma_1\sigma_2} \cup V_z^{\sigma_2}$, we are done by the above arguments. The remaining case is when $p \in H_z^{\sigma_1}$ and $q \in V_z^{\sigma_2}$. Then, any $\pi \in \Pi(p, q)$ intersects both $zh_z^{\sigma_1}$ and $zv_z^{\sigma_2}$. Let $u \in \pi$ be the last point in $\pi \cap zh_z^{\sigma_1}$ from p to q and $w \in \pi$ be the first point in $\pi \cap zv_z^{\sigma_2}$ from p to q . Then, we observe that $u, w \in \partial I_z^{\sigma_1\sigma_2}$. Thus, the subpath of π from u to w is contained in $I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}$ by the lemma when $A = I_z^{\sigma_1\sigma_2} \cup \partial I_z^{\sigma_1\sigma_2}$. On the other hand, the subpath of π from p to u lies in $H_z^{\sigma_1}$ as $p, u \in H_z^{\sigma_1}$, while the subpath of π from w to q lies in $V_z^{\sigma_2}$ as $p, u \in V_z^{\sigma_2}$. This implies that π is contained in $P_z^{\sigma_1\sigma_2}$. This proves the lemma for $A = P_z^{\sigma_1\sigma_2}$. $\square$

Lemma 3.5 *Let $p, q, z \in P$ be any three points. Then, $d(p, q) = d(p, z) + d(z, q)$ if and only if $p \in P_z^{\sigma_1\sigma_2}$ and $q \in P_z^{\sigma'_1\sigma'_2}$ for some $\sigma_1, \sigma_2 \in \{+, -\}$.*

Proof If $p = z$ or $q = z$, then the lemma trivially follows since $z \in P_z^{\sigma_1\sigma_2}$ for any $\sigma_1, \sigma_2 \in \{+, -\}$. So, we assume that $z \notin \{p, q\}$.

We first show the following claim:

(*) *For any $p, q, z \in P$, $d(p, q) = d(p, z) + d(z, q)$ if and only if any $\pi \in \Pi(p, q)$ intersects both the horizontal and vertical chords $h_z^-h_z^+$ and $v_z^-v_z^+$ through z .*

If $\pi \in \Pi(p, q)$ intersects both $h_z^-h_z^+$ and $v_z^-v_z^+$, then let u and u' be any point in the intersection $\pi \cap h_z^-h_z^+$ and $\pi \cap v_z^-v_z^+$, respectively. Then, the subpath of π between u and u' can be replaced by $uz \cup u'z$ by Lemma 2.1. Thus, the new path obtained by this replacement is another shortest path from p to q that passes through z , so we have $d(p, q) = d(p, z) + d(z, q)$. On the other hand, suppose that $d(p, q) = d(p, z) + d(z, q)$. Then, there exists a shortest path $\pi \in \Pi(p, q)$ that passes through z . Since $z = h_z^-h_z^+ \cap v_z^-v_z^+$, this path intersects both $h_z^-h_z^+$ and $v_z^-v_z^+$.

We make use of the above claim (*) to prove the lemma. In the following, we distinguish two cases: for some $\sigma_1, \sigma_2, \sigma'_1, \sigma'_2 \in \{+, -\}$, (1) one of p and q lies in $H_z^{\sigma_1}$ or in $V_z^{\sigma_2}$, or (2) $p \in I_z^{\sigma_1\sigma_2}$ and $q \in I_z^{\sigma'_1\sigma'_2}$.

We first consider the former case and assume that $p \in H_z^-$ without loss of generality. We will show that $d(p, q) = d(p, z) + d(q, z)$ if $q \in P_z^{+-} \cup P_z^{++}$, while $d(p, q) < d(p, z) + d(q, z)$ if $q \in I_z^{--} \cup I_z^{-+} \cup H_z^-$, which implies the lemma since $H_z^- = P_z^{--} \cap P_z^{-+}$. If $q \in I_z^{--} \cup I_z^{-+} \cup H_z^-$, then Lemma 3.4 implies that every shortest path $\pi \in \Pi(p, q)$ is contained in $I_z^{--} \cup I_z^{-+} \cup H_z^-$, so π does not intersect the vertical chord $v_z^-v_z^+$ through z . Thus, we have $d(p, q) < d(p, z) + d(z, q)$ by our claim (*).

Now, consider the case where $q \in P_z^{+-} \cup P_z^{++}$. In this case, we distinguish three cases according to H_z^- . By Lemma 3.2 and Corollary 3.3, either (a) $H_z^- = zh_z^-$, (b) there is a vertex u of P such that for any $p \in H_z^- \setminus zh_z^-$, every shortest path from p

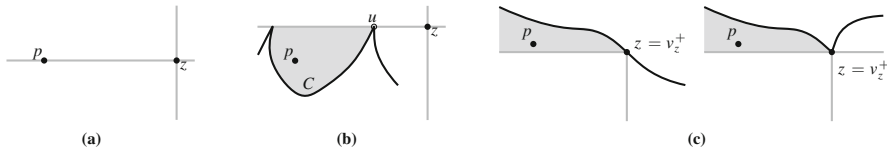


Fig. 3 Illustration to the proof of Lemma 3.5. Note that shaded regions belong to H_z^-

to z passes through u , or (c) $z \in \{v_z^-, v_z^+\}$. See Fig. 3 for an illustration. We show that for any $q \in P_z^{+-} \cup P_z^{++}$, any shortest path $\pi \in \Pi(p, q)$ must intersect the horizontal chord $h_z^- h_z^+$. The case (a) is clear as any $p \in H_z^-$ lies on $h_z^- h_z^+$. In case (b), if $p \in h_z^- h_z^+$, then we are done. Otherwise, p lies in a connected component C of $C_z = P \setminus (h_z^- h_z^+ \cup v_z^- v_z^+)$. Since $q \notin C$ and C is bounded by $h_z^- h_z^+$, the claim is true. Finally, in case (c), assume without loss of generality that $z = v_z^+$. If $p \in h_z^- h_z^+$ or p lies in a connected component of C_z that does not contain z in its boundary, then we are done as in cases (a) and (b). The remaining case is when p lies in a connected component C whose boundary contains z , as illustrated in Fig. 3(c). Then, C lies above $h_z^- h_z^+$. Since the upward half-chord is degenerate to z , any path from p to $q \in P_z^{+-} \cup P_z^{++}$ must intersect $h_z^- h_z^+$.

On the other hand, if $q \in I_z^{++} \cup I_z^{+-} \cup H_z^+$, then any path from p to q must intersect the vertical chord $v_z^- v_z^+$, so we have $d(p, q) = d(p, z) + d(z, q)$ in this case by claim (*). Otherwise, we have $q \in V_z^- \cup V_z^+$. Without loss of generality, we consider the case of $q \in V_z^-$. Then, as done for H_z^- , we have three situations: (a') $V_z^- = zv_z^-$, (b') there is a vertex u' of P such that for any $q \in V_z^- \setminus zv_z^-$, every shortest path from q to z passes through u' , or (c') $z \in \{h_z^-, h_z^+\}$. By a similar argument as above, in cases (a') and (b'), we can show that any shortest path $\pi \in \Pi(p, q)$ must intersect the vertical chord. For case (c'), we must have $z = h_z^+$ since $p \in H_z^-$ and $p \neq z$. This implies that q lies to the right of $v_z^- v_z^+$. Since p lies to the left of $v_z^- v_z^+$, this implies that any $\pi \in \Pi(p, q)$ must intersect the vertical chord. Hence, in either case, any $\pi \in \Pi(p, q)$ intersects both the horizontal and vertical chords through z , so we have $d(p, q) = d(p, z) + d(z, q)$ by our claim (*).

We then turn to case (2) where $p \in I_z^{\sigma_1 \sigma_2}$ and $q \in I_z^{\sigma'_1 \sigma'_2}$ for some $\sigma_1, \sigma_2, \sigma'_1, \sigma'_2 \in \{+, -\}$. Without loss of generality, we assume that $\sigma_1 = \sigma_2 = -$. By symmetry, we consider three cases: $q \in I_z^{++}$, $q \in I_z^{--}$, or $q \in I_z^{-+}$. If $p \in I_z^{--}$ and $q \in I_z^{++}$, then any path from p to q must intersect both the horizontal and vertical chords through z , so we have $d(p, q) = d(p, z) + d(z, q)$ by our claim (*). Otherwise, assume that $p \in I_z^{--}$ and $q \in I_z^{--} \cup I_z^{-+}$. Consider the shortest paths $\pi_2(p, z)$ and $\pi_2(q, z)$, and let u and u' be the last vertices of $\pi_2(p, z)$ and $\pi_2(q, z)$, respectively. Then, observe that $u \in I_z^{--}$ and $u' \in I_z^{--} \cup I_z^{-+}$ by Lemma 3.4. Thus, both uz and $u'z$, except z , lie in the left side of the vertical chord $v_z^- v_z^+$ through z . This implies the existence of a vertical line ℓ such that ℓ crosses both uz and $u'z$ at points w and w' , respectively, and $ww' \subset P$. Then, we obtain a path π from p to q by concatenating $\pi_2(p, w)$, ww' and $\pi_2(w', q)$. Observe that

$$|\pi| < |\pi_2(p, z)| + |\pi_2(q, z)| = d(p, z) + d(q, z)$$

by Lemma 2.1. This implies that $d(p, q) \leq |\pi| < d(p, z) + d(z, q)$, completing the proof. $\square$

4 L_1 Geodesic Farthest Neighbors and Extreme Points

Let $S \subseteq P$ be a nonempty, compact subset of P . We are interested in farthest neighbors of each $p \in P$ from S . For each $p \in P$, let $\Phi_S(p) := \max_{q \in S} d(p, q)$. This is well defined since S is a compact set. We call such a $q \in S$ with $d(p, q) = \Phi_S(p)$ an L_1 geodesic farthest neighbor of p from S , or shortly a farthest neighbor of p when there is no confusion. There can be several farthest neighbors of $p \in P$ from S . We denote by $F_S(p) \subseteq S$ the set of all farthest neighbors of p from S . In order to pick a representative among them, we impose a total order $<$ on S , such as the lexicographical order. We then define $f_S(p) \in F_S(p)$ to be the least with respect to the order $<$ among the farthest neighbors of p in $F_S(p)$. We call $q \in S$ an $(L_1 \text{ geodesic})$ extreme point of S if $q = f_S(p)$ for some $p \in P$.

There are two fundamental quantities defined by farthest neighbors in P : the $(L_1 \text{ geodesic})$ diameter $\text{diam}(S) := \max_{q \in S} \Phi_S(q)$ and the $(L_1 \text{ geodesic})$ radius $\text{rad}(S) := \min_{p \in P} \Phi_S(p)$ of S . The diameter and radius of S are well defined since P and S are compact sets. A pair of points $q, q' \in S$ is called *diametral* if $d(q, q') = \text{diam}(S)$, while a point $c \in P$ is called a $(L_1 \text{ geodesic})$ center of S if $\Phi_S(c) = \text{rad}(S)$. Let $\text{cen}(S)$ be the set of all centers $c \in P$ of S .

In this section, we fully reveal the behavior of the L_1 farthest neighbors and extreme points of any compact set S in P , and finally prove the following theorem.

Theorem 4.1 *In a simple polygon P , there are at most four extreme points of any compact subset $S \subseteq P$ with respect to the L_1 geodesic distance.*

In order to prove Theorem 4.1, we consider farthest neighbors constrained in regions. For $\sigma_1, \sigma_2 \in \{+, -\}$, define $f_S^{\sigma_1 \sigma_2}(p)$ to be the farthest neighbor of p from $S \cap P_p^{\sigma_1 \sigma_2}$ that is the least with respect to $<$. In the case where $S \cap P_p^{\sigma_1 \sigma_2} = \emptyset$, $f_S^{\sigma_1 \sigma_2}(p)$ is undefined. Then observe that $f_S(p)$ is the farthest one that is the least with respect to $<$ among the four candidates $f_S^{++}(p)$, $f_S^{-+}(p)$, $f_S^{--}(p)$, and $f_S^{+-}(p)$.

We first gather some useful properties of the constrained farthest neighbors.

Lemma 4.2 *Given any $p \in P$, suppose that $f_S^{\sigma_1 \sigma_2}(p) \in F_S(p)$ for $\sigma_1, \sigma_2 \in \{+, -\}$. Then, $f_S^{\sigma_1 \sigma_2}(p) \in F_S(p')$ for any $p' \in P_p^{\overline{\sigma_1 \sigma_2}}$, and $f_S(p') = f_S^{\sigma_1 \sigma_2}(p)$ for any $p' \in I_p^{\overline{\sigma_1 \sigma_2}}$. Moreover, if $f_S(p) = f_S^{\sigma_1 \sigma_2}(p)$, then $f_S(p') = f_S^{\sigma_1 \sigma_2}(p)$ for any $p' \in P_p^{\overline{\sigma_1 \sigma_2}}$.*

Proof The lemma consists of three statements. We prove them separately in the following.

Note that $\max_{q \in S} d(p, q) = \Phi_S(p) = d(p, f_S^{\sigma_1 \sigma_2}(p))$ since $f_S^{\sigma_1 \sigma_2}(p) \in F_S(p)$. For any $p' \in P_p^{\overline{\sigma_1 \sigma_2}}$ and any $q \in S$, we have

$$d(p', q) \leq d(p', p) + d(p, q) \leq d(p', p) + d(p, f_S^{\sigma_1 \sigma_2}(p)) = d(p', f_S^{\sigma_1 \sigma_2}(p))$$

by Lemma 3.5, since $p' \in P_p^{\overline{\sigma_1 \sigma_2}}$ and $f_S^{\sigma_1 \sigma_2}(p) \in P_p^{\sigma_1 \sigma_2}$. Thus, we have $f_S^{\sigma_1 \sigma_2}(p) \in F_S(p')$ for any $p' \in P_p^{\overline{\sigma_1 \sigma_2}}$. This proves the first statement.

We then turn to the third statement, so assume in addition that $f_S(p) = f_S^{\sigma_1\sigma_2}(p)$. For any $p' \in P_p^{\overline{\sigma_1\sigma_2}}$, suppose to the contrary that $f_S(p') \neq f_S(p) = f_S^{\sigma_1\sigma_2}(p)$. Then, we must have $d(p', f_S(p')) = d(p', f_S(p))$ and $f_S(p') < f_S(p)$, by the above argument. Since $d(p', f_S(p)) = d(p', p) + d(p, f_S(p))$ and $d(p', f_S(p')) \leq d(p', p) + d(p, f_S(p'))$, this also implies that $f_S(p')$ is another farthest neighbor of p from S , that is, $f_S(p') \in F_S(p)$. But this leads to a contradiction to the assumption that $f_S(p)$ is the least among the farthest neighbors in $F_S(p)$ of p from S with respect to $<$. Hence, we have $f_S(p') = f_S(p) = f_S^{\sigma_1\sigma_2}(p)$.

In order to prove the second statement, we let $p' \in I_p^{\overline{\sigma_1\sigma_2}}$. Then, for any $q \in S \cap P_p^{\sigma_1\sigma_2}$, we have the equality $d(p', q) = d(p', p) + d(p, q)$, while for any $q \in S \setminus P_p^{\sigma_1\sigma_2}$, we have the strict inequality $d(p', q) < d(p', p) + d(p, q)$ by Lemma 3.5. This implies that $F_S(p') = F_S(p) \cap P_p^{\sigma_1\sigma_2}$. Since $f_S^{\sigma_1\sigma_2}(p) \in F_S(p)$ is the least among those in $F_S(p) \cap P_p^{\sigma_1\sigma_2}$, we conclude that $F_S(p') = f_S^{\sigma_1\sigma_2}(p)$. $\square$

Lemma 4.3 *For any $p \in P$, let $\pi \in \Pi(p, f_S(p))$ and $z \in \pi$. Then, for any $\sigma_1, \sigma_2 \in \{+, -\}$ with $p \in P_z^{\sigma_1\sigma_2}$ and $f_S(p) \in P_z^{\overline{\sigma_1\sigma_2}}$, it holds that $f_S(p) = f_S^{\sigma_1\sigma_2}(z)$.*

Proof Fix any $z \in \pi$ on any shortest path $\pi \in \Pi(p, f_S(p))$ from p to $f_S(p)$. Let $\sigma_1, \sigma_2 \in \{+, -\}$ be such that $p \in P_z^{\sigma_1\sigma_2}$ and $f_S(p) \in P_z^{\overline{\sigma_1\sigma_2}}$. Without loss of generality, we assume that $\sigma_1 = \sigma_2 = -$, so $f_S(p) \in P_z^{++}$.

Let $q = f_S^{++}(z) \in S$. Suppose to the contrary that $q \neq f_S(p)$. Since $p \in P_z^{--}$, we have $d(p, q) = d(p, z) + d(z, q)$ by Lemma 3.5. By definition, q is the least one among those $q' \in S \cap P_z^{++}$ that maximizes the distance $d(z, q')$. Since $f_S(p) \in P_z^{++}$, we also have

$$d(p, f_S(p)) = d(p, z) + d(z, f_S(p)) \leq d(p, z) + d(z, q) = d(p, q)$$

by Lemma 3.5. This implies that $d(z, f_S(p)) = d(z, q)$ since $f_S(p) \in F_S(p)$. That is, both q and $f_S(p)$ are farthest neighbors of z from $S \cap P_z^{++}$, and simultaneously are farthest neighbors of p from S . If $q < f_S(p)$, then we must have $f_S(p) = f_S^{++}(z) = q$; otherwise, if $f_S(p) < q$, then we must have $f_S(p) = f_S^{++}(z)$. In either case, we must have $f_S(p) = q$, a contradiction. $\square$

Note that such $\sigma_1, \sigma_2 \in \{+, -\}$ with $p \in P_z^{\sigma_1\sigma_2}$ and $f_S(p) \in P_z^{\overline{\sigma_1\sigma_2}}$ in Lemma 4.3 always exist by Lemma 3.5 since z is a point on a shortest path from p to $f_S(p)$, so $d(p, f_S(p)) = d(p, z) + d(z, f_S(p))$.

Next, we pick a center $c \in \text{cen}(S)$ of S and consider the set $F_S(c)$ of its farthest neighbors. Since c is a center, we have $d(c, \chi) = \Phi_S(c) = \text{rad}(S)$ for any $\chi \in F_S(c)$.

Lemma 4.4 *For any $c \in \text{cen}(S)$, there exist $\chi_1, \chi_2 \in F_S(c)$ satisfying the following: (i) $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$, and (ii) $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$.*

Proof If S consists of a single point q , then the lemma is trivially true since we have $c = q$ and further observe that (i) $d(q, q) = d(q, c) + d(c, q) = 0$ and (ii) $f_S(q) = q$. So, in the following, we assume that S consists of at least two points. Then, we have $|F_S(c)| \geq 2$.

We first verify the existence of $\chi_1, \chi_2 \in F_S(c)$ with (i) $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$. Recall Lemma 3.5 that we have $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$ if and only if $\chi_1 \in P_c^{\sigma_1\sigma_2}$ and $\chi_2 \in P_c^{\overline{\sigma_1}\overline{\sigma_2}}$ for some $\sigma_1, \sigma_2 \in \{+, -\}$. Thus, if there exist $\chi_1, \chi_2 \in F_S(c)$ such that $\chi_1 \in P_c^{\sigma_1\sigma_2}$ and $\chi_2 \in P_c^{\overline{\sigma_1}\overline{\sigma_2}}$ for some $\sigma_1, \sigma_2 \in \{+, -\}$, then we are done.

For contradiction, suppose that there are no two such farthest neighbors of c in $F_S(c)$. Then, we claim that all those in $F_S(c)$ are contained in either

$$I_c^{\sigma_1\sigma_2} \cup V_c^{\sigma_2} \cup I_c^{\overline{\sigma_1}\overline{\sigma_2}} \quad \text{or} \quad I_c^{\sigma_1\sigma_2} \cup H_c^{\sigma_1} \cup I_c^{\sigma_1\overline{\sigma_2}},$$

for some $\sigma_1, \sigma_2 \in \{+, -\}$.

In order to see the claim, assume that there is $\chi \in F_S(c)$ with $\chi \in P_c^{\sigma_1\sigma_2}$. Then, no $\chi' \in F_S(c)$ is contained in $P_c^{\overline{\sigma_1}\overline{\sigma_2}}$, that is, every $\chi' \in F_S(c)$ is contained in $I_c^{\sigma_1\sigma_2}, H_c^{\sigma_1}, I_c^{\sigma_1\sigma_2}, V_c^{\sigma_2}$, or $I_c^{\overline{\sigma_1}\overline{\sigma_2}}$. Recall that $P_c^{\sigma_1\sigma_2} = I_c^{\sigma_1\sigma_2} \cup H_c^{\sigma_1} \cup V_c^{\sigma_2}$. If there is $\chi' \in F_S(c)$ with $\chi' \in I_c^{\sigma_1\overline{\sigma_2}} \cup H_c^{\sigma_1}$, then for the same reason, there is no $\chi'' \in F_S(c)$ with $\chi'' \in I_c^{\overline{\sigma_1}\overline{\sigma_2}} \cup V_c^{\sigma_2}$. This implies the claim.

Thus, we have $F_S(c) \subset I_c^{\sigma_1\sigma_2} \cup V_c^{\sigma_2} \cup I_c^{\overline{\sigma_1}\overline{\sigma_2}}$, or $F_S(c) \subset I_c^{\sigma_1\sigma_2} \cup H_c^{\sigma_1} \cup I_c^{\sigma_1\overline{\sigma_2}}$, for some $\sigma_1, \sigma_2 \in \{+, -\}$. We only consider the former case, as the other case is symmetric. Also, without loss of generality, assume that $\sigma_1 = \sigma_2 = -$, so $F_S(c) \subset I_c^{--} \cup V_c^- \cup I_c^{-+}$. Recall the definitions that for any $p \in V_c^-$ with $p \neq c$, there is a shortest path $\pi \in \Pi(p, c)$ that chooses the downward half-chord cv_c^- from c , that is, $\pi \cap cv_c^-$ contains a point other than c . Also, for any $p \in I_c^{-+}$, there are two shortest paths $\pi, \pi' \in \Pi(p, c)$ such that π chooses the leftward half-chord from c and π' chooses the downward half-chord from c . This implies that for any $p \in I_c^{--} \cup V_c^- \cup I_c^{-+}$, there is a shortest path $\pi \in \Pi(p, c)$ that chooses the downward half-chord from c .

Now, consider all farthest neighbors $\chi \in F_S(c)$ of c . Note that $\chi \neq c$ for any $\chi \in F_S(c)$ since $|F_S(c)| \geq 2$. Since $\chi \in I_c^{--} \cup V_c^- \cup I_c^{-+}$, there is a shortest path $\pi_\chi \in \Pi(\chi, c)$ that chooses the downward half-chord from c . Then, we can choose a point $z \in cv_c^-$ with $z \neq c$ that is arbitrarily close to z and is passed through by the path π_χ for all $\chi \in F_S(c)$. It is not difficult to see that $d(z, \chi) < d(c, \chi)$ for any $\chi \in F_S(c)$. On the other hand, by our choice of z , we still have that for any $q \in S \setminus F_S(c)$ and any $\chi \in F_S(c)$, $d(z, \chi) > d(z, q)$. This implies that $\Phi_S(z) < \Phi_S(c) = \text{rad}(S)$, a contradiction. Hence, there exist $\chi_1, \chi_2 \in F_S(c)$ such that $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$.

Note that for $\chi_1, \chi_2 \in F_S(c)$, if $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$, then (χ_1, χ_2) is a diametral pair, that is, $d(\chi_1, \chi_2) = \text{diam}(S)$, since $d(\chi_1, c) + d(c, \chi_2) = 2\text{rad}(S)$ and $d(q, q') \leq d(q, c) + d(c, q') \leq 2\text{rad}(S)$ for any $q, q' \in S$. This implies that $\text{diam}(S) = 2\text{rad}(S)$. Now, consider $D := \{q \in S \mid \Phi_S(q) = \text{diam}(S)\}$. Then, observe that $D \subseteq F_S(c)$ since $\Phi_S(q) < \text{diam}(S) = 2\text{rad}(S)$ if $d(c, q) < \text{rad}(S)$.

Now, we consider a directed graph $G = (D, E)$ such that there is an arc $(\chi_1, \chi_2) \in E$ if and only if $f_S(\chi_1) = \chi_2$. Note that $|E| = |D|$ since $f_S(\chi_1) \in D$ is unique. In the following, we show that there is no directed cycle of length three or longer in G .

Suppose that there is a cycle of length k in G with $k \geq 3$. Let $\chi_0, \dots, \chi_{k-1} \in D$ be those in the cycle in this order such that $(\chi_i, \chi_{i+1}) \in E$ for each $i = 0, \dots, k-1$

where the indices are taken by modulo k . Thus, we have $d(\chi_i, \chi_{i+1}) = d(\chi_{i-1}, \chi_i) = \text{diam}(S)$ and $f_S(\chi_i) = \chi_{i+1}$. This implies that $\chi_{i+1} \prec \chi_{i-1}$ for any $i = 0, \dots, k-1$. If k is odd, then we have a chain

$$\chi_{k-1} \prec \dots \prec \chi_2 \prec \chi_0 \prec \chi_{k-2} \prec \dots \prec \chi_3 \prec \chi_1 \prec \chi_{k-1},$$

leading to a contradiction. Otherwise, if k is even, then we also have

$$\chi_{k-1} \prec \dots \prec \chi_3 \prec \chi_1 \prec \chi_{k-1},$$

leading to a contradiction.

Suppose that G has no two arcs (χ_1, χ_2) and (χ_2, χ_1) for some $\chi_1, \chi_2 \in D$. Then, G is a directed acyclic graph and thus there exists a topological order on D with respect to G . This implies a last element χ in any topological order on D such that χ has no out-going arc of the form (χ, χ') in G . This is a contradiction since every $\chi \in D$ has exactly out-degree one since $f_S(\chi) \in D$ is uniquely determined.

Consequently, there are $\chi_1, \chi_2 \in D$ such that $(\chi_1, \chi_2), (\chi_2, \chi_1) \in E$. Since $D \subseteq F_S(c)$, this proves the existence of $\chi_1, \chi_2 \in F_S(c)$ with (i) $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$ and (ii) $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$. $\square$

4.1 Proof of Theorem 4.1

Now, we give a proof of Theorem 4.1. The case where S consists of at most one point is trivial, so we assume that S consists of more than one point.

We first fix any two farthest neighbors $\chi_1, \chi_2 \in F_S(c)$ of c with the property of Lemma 4.4. Note that χ_1 and χ_2 are extreme points of S . Since $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$, we have $\chi_1 \in P_c^{\sigma_1 \sigma_2}$ and $\chi_2 \in P_c^{\overline{\sigma_1} \overline{\sigma_2}}$ for some $\sigma_1, \sigma_2 \in \{+, -\}$ by Lemma 3.5. Without loss of generality, we assume that $\sigma_1 = \sigma_2 = -$, so $\chi_1 \in P_c^{--}$ and $\chi_2 \in P_c^{++}$.

Consider the path π from χ_1 to χ_2 obtained by concatenating two unique Euclidean shortest paths $\pi_2(\chi_1, c)$ and $\pi_2(c, \chi_2)$, that is, $\pi = \pi_2(\chi_1, c) \cup \pi_2(c, \chi_2)$. Since $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$, π is a shortest path from χ_1 to χ_2 , that is, $\pi \in \Pi(\chi_1, \chi_2)$. Then, by Lemma 4.3, we have $\chi_1 = f_S^{--}(c)$ and $\chi_2 = f_S^{++}(c)$, as $c \in \pi$. This further implies that $f_S(p) = \chi_2$ for any $p \in I_c^{--}$, and $f_S(p) = \chi_1$ for any $p \in I_c^{++}$ by Lemma 4.2 since $\chi_1, \chi_2 \in F_S(c)$.

We will need the following lemma, which rephrases the Ordering Lemma by Aronov et al. [2]. Note that every extreme point of S appears on the boundary of the relative convex hull $\text{rconv}(S)$ of S .

Lemma 4.5 (Aronov et al. [2]) *Suppose that there are three distinct extreme points χ_1, χ_2, χ_3 of S in the counter-clockwise order along $\partial \text{rconv}(S)$. Let $p_i \in \partial P$ be a point on the boundary of P such that $f_S(p_i) = \chi_i$ for each $i \in \{1, 2, 3\}$. Then, p_1, p_2, p_3 appear in this order along ∂P in the counter-clockwise direction.*

Consider the extension of the last segment of $\pi_2(c, \chi_i)$ for each $i \in \{1, 2\}$ until it hits the first boundary point $\hat{\chi}_i \in \partial P$. Let $\hat{\pi}$ be the shortest path from $\hat{\chi}_1$ to $\hat{\chi}_2$

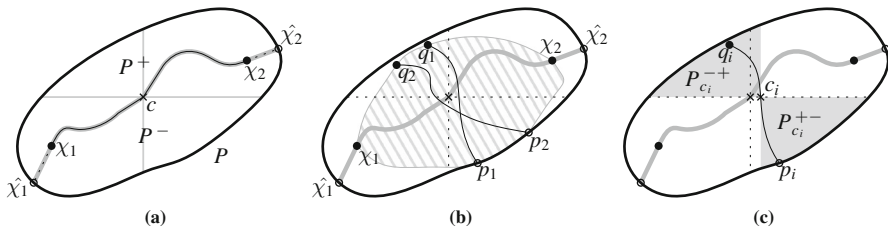


Fig. 4 (a) The path $\hat{\pi}$ through $c \in \text{cen}(S)$ partitions P into P^+ and P^- . (b) Illustration to Claims 4.6 and 4.7. Shaded region represents $\text{rconv}(S \cup \{p_1, p_2\})$. (c) Illustration to Claim 4.8

obtained by these extensions from π ; that is, $\hat{\pi} = \pi_2(\hat{\chi}_1, c) \cup \pi_2(c, \hat{\chi}_2)$. Note that $f_S(\hat{\chi}_1) = \chi_2$ and $f_S(\hat{\chi}_2) = \chi_1$ by Lemma 4.2. The path $\hat{\pi}$ partitions P into two parts P^- and P^+ , where ∂P^- consists of $\hat{\pi}$ and the chain along ∂P from $\hat{\chi}_1$ to $\hat{\chi}_2$ in the counter-clockwise direction, and ∂P^+ consists of $\hat{\pi}$ and the chain along ∂P from $\hat{\chi}_2$ to $\hat{\chi}_1$ in the counter-clockwise direction. See Fig. 4(a) for an illustration. In the following, we show that there is at most one more extreme point of S , other than χ_1 and χ_2 , in each of P^- and P^+ . Recall that an extreme point of S is $q \in S$ such that $q = f_S(p)$ for some $p \in P$.

Suppose to the contrary that there are two extreme points q_1, q_2 of S such that $q_1, q_2 \in P^+$ and the four points χ_1, χ_2, q_1, q_2 are all distinct. Then there exist two boundary points $p_1, p_2 \in \partial P$ such that $f_S(p_1) = q_1$ and $f_S(p_2) = q_2$. Such boundary points p_1, p_2 are guaranteed to exist by Lemma 4.2. Our proof will be done by a contradiction based on the following four claims.

Claim 4.6 Both p_1 and p_2 lie in I_c^{+-} .

Proof Note that $p_1, p_2 \in P_c^{+-} \cup P_c^{-+}$ since for any $p \in I_c^{++} \cup I_c^{--}$, $f_S(p)$ is either χ_1 or χ_2 , as discussed above. Suppose that $p_i \in P_c^{-+}$ for $i \in \{1, 2\}$. Then, we meet $\hat{\chi}_2, p_i, \hat{\chi}_1$ in this order when walking along ∂P in the counter-clockwise direction. On the other hand, when we walk along the boundary $\partial \text{rconv}(S)$ of the relative convex hull of S in the counter-clockwise direction, we meet χ_2, q_i, χ_1 since $q_i \in P^+$. This contradicts Lemma 4.5 since $f_S(\hat{\chi}_2) = \chi_1$ and $f_S(\hat{\chi}_1) = \chi_2$. Hence, $p_1, p_2 \in P_c^{+-}$.

In order to show that indeed $p_1, p_2 \in I_c^{+-}$, suppose to the contrary that $p_1 \in H_c^+$ or $p_1 \in V_c^-$. If $p_1 \in H_c^+$, then we have $\chi_1 \in F_S(p_1)$ by Lemma 4.2 since $\chi_1 = f_S^{--}(c)$ and $p_1 \in P_c^{+-}$. Thus, $d(p_1, q_1) = d(p_1, \chi_1)$ and further $d(c, q_1) = d(c, \chi_1)$ since $d(p_1, q_1) \leq d(p_1, c) + d(c, q_1)$ and $\chi_1 \in F_S(c)$. As $q_1 = f_S(p_1)$ and $q_1 \neq \chi_1$, we must have $q_1 = f_S^{--}(c) \in F_S(c)$. By the same argument, in case of $p_1 \in V_c^-$, we also have $q_1 = f_S^{--}(c) \in F_S(c)$. Consequently, if $p_1 \in H_c^+$ or $p_1 \in V_c^-$, then $q_1 = f_S^{--}(c)$ and $q_1 \in F_S(c)$.

Now, consider p_2 . If $p_2 \in H_c^+$ or $p_2 \in V_c^-$, then by the same argument, we have $q_2 = f_S^{--}(c)$, a contradiction to the assumption that $q_1 \neq q_2$. Otherwise, if $q_2 \in I_c^{+-}$, then we still have $q_1 \in F_S(p_2)$ since for any $q \in S$, it holds that

$$d(p_2, q) \leq d(p_2, c) + d(c, q) \leq d(p_2, c) + d(c, q_1) = d(p_2, q_1)$$

by Lemma 3.5. On the other hand, for any $q \in S \setminus P_c^{+-}$, we have a strict inequality

$$d(p_2, q) < d(p_2, c) + d(c, q) \leq d(p_2, c) + d(c, q_1) = d(p_2, q_1) = \Phi_S(p_2)$$

again by Lemma 3.5. Since $q_2 = f_S(p_2)$, we have $d(p_2, q_2) = d(p_2, q_1) = \Phi_S(p_2)$, and thus $q_2 \in P_c^{+-}$. This implies that we must have $q_2 = f_S(p_2) = f_S^{+-}(c)$, a contradiction. Hence, both p_1 and p_2 lie in I_c^{+-} . $\square$

In the following, we assume that the four points χ_2, q_1, q_2, χ_1 appear in this order along $\partial \text{rconv}(S)$ in the counter-clockwise direction. Then, Lemma 4.5 implies the following.

Claim 4.7 *The four points p_1, p_2, q_1, q_2 appear in this order along $\partial \text{rconv}(S \cup \{p_1, p_2\})$.*

Proof Consider the relative convex hull $C := \text{rconv}(S \cup \{p_1, p_2\})$ of S and in addition p_1, p_2 . Note that p_1 and p_2 appear on the boundary ∂C of the relative convex hull $C = \text{rconv}(S \cup \{p_1, p_2\})$, since $p_1, p_2 \in \partial P$. On the other hand, q_1 and q_2 also appear on ∂C since $q_1 = f_S(p_1)$ and $q_2 = f_S(p_2)$.

Now, consider the order of four points χ_1, χ_2, q_1, q_2 along $\partial \text{rconv}(S)$. Since $q_1, q_2 \in P^+$, $\chi_1 \in P_c^{--}$ and $\chi_2 \in P_c^{++}$, there are two possibilities: $(\chi_2, q_1, q_2, \chi_1)$ or $(\chi_2, q_2, q_1, \chi_1)$. Since the four points χ_2, q_1, q_2, χ_1 appear in this order along $\partial \text{rconv}(S)$ in the counter-clockwise direction, the four points $\hat{\chi}_1, p_1, p_2, \hat{\chi}_2$ appear in this order along ∂C in the counter-clockwise direction by Lemma 4.5 and Claim 4.6. This implies that the six points $\chi_1, p_1, p_2, \chi_2, q_1, q_2$ appear in this order along ∂C in the counter-clockwise direction. See Fig. 4 (b) for an illustration. This implies the claim. $\square$

See Fig. 4 (b) for an illustration to the above two claims. Note that Claim 4.6 implies that $p_1, p_2 \in P^-$ as $I_c^{+-} \subset P^-$, so any shortest path from p_i to q_i crosses $\hat{\pi}$. On the other hand, Claim 4.7 implies that $\pi_2(p_1, q_1)$ and $\pi_2(p_2, q_2)$ cross each other.

Let $\beta := \partial I_c^{+-} \setminus \partial P$. Note that β is a subset of the union of the rightward half-chord ch_c^+ and the downward half-chord cv_c^- from c . By Claim 4.6, the paths $\pi_2(p_1, q_1)$ and $\pi_2(p_2, q_2)$ must cross over β as $p_1, p_2 \in I_c^{+-}$ and $q_1, q_2 \in P^+$. For each $i \in \{1, 2\}$, let c_i be the first intersection point of $\pi_2(p_i, q_i) \cap \beta$ when walking from p_i to q_i along $\pi_2(p_i, q_i)$. We then observe the following.

Claim 4.8 *For $i \in \{1, 2\}$, we have $p_i \in P_{c_i}^{+-}$ and $q_i \in P_{c_i}^{-+}$.*

Proof The point c_i lies either on ch_c^+ or on cv_c^- . We handle the former case only, as the other case can be handled in a symmetrical way. Then, we have $q_i \in P_{c_i}^{-+} \cup P_{c_i}^{++}$ and $p_i \in P_{c_i}^{--} \cup P_{c_i}^{+-}$.

We first show that if $q_i \in P_{c_i}^{++}$, then $\chi_2 \in P_{c_i}^{++}$. Suppose that $q_i \in P_{c_i}^{-+}$, but $\chi_2 \notin P_{c_i}^{++}$. See Fig. 5 (a) for an illustration. By Claim 4.6, there is an intersection point y between $\hat{\pi}$ and $\pi_2(p_i, q_i)$. Observe that $\pi_2(p_i, c_i)$ is contained in I_c^{+-} , except c_i , and $\pi_2(c_i, q_i)$ is contained in $P_{c_i}^{++}$ by Lemma 3.4 and Claim 4.6. Also, recall that

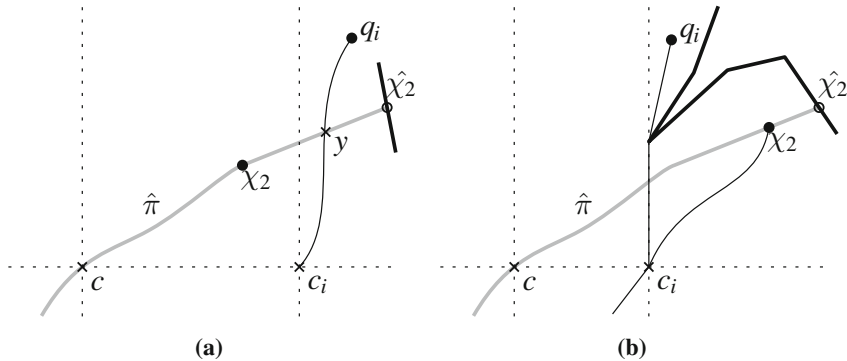


Fig. 5 Illustration to the proof of Claim 4.8. (a) When $q_i \in P_{c_i}^{++}$ but $\chi_2 \notin P_{c_i}^{++}$. (b) When $p_i \notin P_{c_i}^{+-}$

$\hat{\pi} = \pi_2(\hat{\chi}_1, c) \cup \pi_2(c, \hat{\chi}_2)$, $\chi_1 \in P_c^{--}$, and $\chi_2 \in P_c^{++}$. Thus, again by Lemma 3.4, $\hat{\pi} \cap I_c^{+-} = \emptyset$. This implies that $y \in P_{c_i}^{++}$. We then have

$$d(c, \chi_2) < d(c, y) = d(c, c_i) + d(c_i, y) \leq d(c, c_i) + d(c_i, q_i) = d(c, q_i),$$

a contradiction to $f_S^{++}(c) = \chi_2$.

Next, we show that $q_i \in P_{c_i}^{+-}$. Suppose to the contrary that $q_i \notin P_{c_i}^{+-}$, so $q_i \in I_{c_i}^{++} \cup H_{c_i}^+$. Then, by Lemma 3.5, we have $p_i \in P_{c_i}^{--}$ since, otherwise, if $p_i \notin P_{c_i}^{--}$, then we have the inequality $d(p_i, q_i) < d(p_i, c_i) + d(c_i, q_i)$. Thus, Lemma 4.3 implies that $q_i = f_S^{++}(c_i)$. Since $I_{c_i}^{++} \cup H_{c_i}^+ \subset P_{c_i}^{++}$, by the above argument, we have $\chi_2 \in P_{c_i}^{++}$. Then, since $q_i = f_S^{++}(c_i)$ and $c \in P_{c_i}^{--}$, Lemma 4.2 implies that $f_S^{++}(c) = q_i$. But this leads to a contradiction since $f_S^{++}(c) = \chi_2$. Hence, $q_i \in P_{c_i}^{+-}$.

Finally, we show that $p_i \in P_{c_i}^{+-}$. Suppose to the contrary that $p_i \notin P_{c_i}^{+-}$. Then, we have $p_i \in I_{c_i}^{--} \cup H_{c_i}^-$. Since $d(p_i, q_i) = d(p_i, c_i) + d(c_i, q_i)$, we have $q_i \in P_{c_i}^{++}$ by Lemma 3.5, so indeed $q_i \in V_{c_i}^+ = P_{c_i}^{+-} \cap P_{c_i}^{++}$. See Fig. 5 (b). By the first argument, we also have $\chi_2 \in P_{c_i}^{++}$. As $p_i \in I_{c_i}^{--} \cup H_{c_i}^- \subset P_{c_i}^{--}$, we have $d(p_i, \chi_2) = d(p_i, c_i) + d(c_i, \chi_2)$ again by Lemma 3.5. Since $q_i = f_S(p_i)$, this implies that $d(c_i, \chi_2) \leq d(c_i, q_i)$. Moreover, since $c \in P_{c_i}^{--}$, we have

$$d(c, \chi_2) = d(c, c_i) + d(c_i, \chi_2) \leq d(c, c_i) + d(c_i, q_i) = d(c, q_i),$$

so we have $d(c, \chi_2) = d(c, q_i)$ and $q_i \in F_S(c)$. Since $f_S^{++}(c) = \chi_2$, we have $\chi_2 < q_i$. On the other hand, as $d(c, q_i) = d(c, c_i) + d(c_i, q_i)$ and $d(c, \chi_2) = d(c, c_i) + d(c_i, \chi_2)$, we have $d(c_i, \chi_2) = d(c_i, q_i)$ and thus $d(p_i, \chi_2) = d(p_i, q_i)$, so $\chi_2 \in F_S(p_i)$. Since $f_S(p_i) = q_i$, we must have $q_i < \chi_2$, a contradiction. $\square$

See Fig. 4 (c) for an illustration to Claim 4.8. Our last claim to prove Theorem 4.1 is the following.

Claim 4.9 *There exists $c' \in P$ such that $p_1, p_2 \in P_{c'}^{+-}$ and $q_1, q_2 \in P_{c'}^{+-}$.*

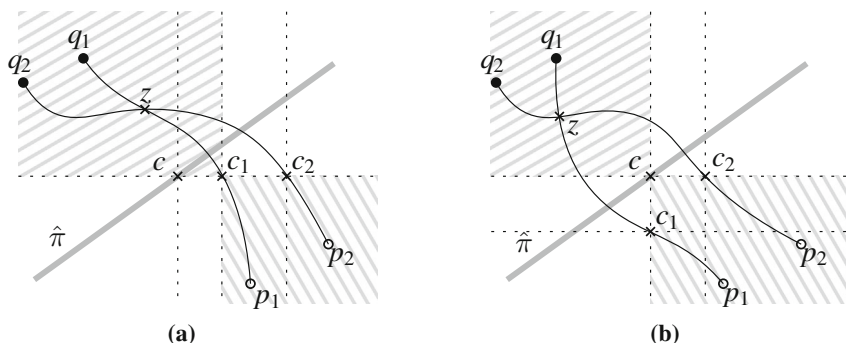


Fig. 6 Illustration to the proof of Claim 4.8 when $z \notin I_c^{+-}$. Shaded regions depict $P_{c'}^{-+}$ and $P_{c'}^{+-}$. (a) If $c_1, c_2 \in ch_c^+$, we have $c_1 \in cc_2$, and take $c' = c_1$. (b) If $c_1 \in cv_c^-$ and $c_2 \in ch_c^+$, we take $c' = c$

Proof Recall that we meet the four points p_1, p_2, q_1, q_2 in this order when walking along $\partial \text{rconv}(S \cup \{p_1, p_2\})$ by Claim 4.7. Thus, $\pi_2(p_1, q_1)$ and $\pi_2(p_2, q_2)$ cross each other. Let $z \in \pi_2(p_1, q_1) \cap \pi_2(p_2, q_2)$ be the first intersection point along the two paths from p_1 and p_2 , respectively, to q_1 and q_2 , respectively. In this proof, we consider two different cases: $z \in I_c^{+-}$ or $z \notin I_c^{+-}$.

Suppose that $z \notin I_c^{+-}$. We define $c' \in P$ to be such that $P_{c'}^{-+} = P_{c_1}^{-+} \cap P_{c_2}^{-+}$. Such c' is uniquely determined as follows. Since $z \notin I_c^{+-}$, observe that the four points p_1, p_2, c_2, c_1 appear in this order along ∂I_c^{+-} . If $c_1, c_2 \in ch_c^+$, then c_1 must lie in between c and c_2 , so we take $c' = c_1$. See Fig. 6 (a). If $c_1, c_2 \in cv_c^-$, then c_2 lies in between c and c_1 , so we have $c' = c_2$. Otherwise, it always holds that $c_1 \in cv_c^-$ and $c_2 \in ch_c^+$ by the order along ∂I_c^{+-} . In this case, we take $c' = c$. See Fig. 6 (b).

We then verify that $p_1, p_2 \in P_{c'}^{+-}$ and $q_1, q_2 \in P_{c'}^{-+}$. By our choice of c' , observe that

$$P_{c'}^{-+} = P_{c_1}^{-+} \cap P_{c_2}^{-+} \text{ and } P_{c_1}^{+-} \cup P_{c_2}^{+-} \subseteq P_{c'}^{+-}.$$

Since $p_i \in P_{c_i}^{+-}$ by Claim 4.8, this already proves that $p_1, p_2 \in P_{c'}^{+-}$. In order to see that $q_1, q_2 \in P_{c'}^{-+}$, observe that $\partial \text{rconv}(S \cup \{p_1, p_2\})$ intersects the three sets

$$P_{c_2}^{-+} \setminus P_{c_1}^{-+}, \quad P_{c_1}^{-+} \cap P_{c_2}^{-+}, \quad P_{c_1}^{-+} \setminus P_{c_2}^{-+}$$

in this order along $\partial \text{rconv}(S \cup \{p_1, p_2\})$, if nonempty. Thus, if $q_1 \in P_{c_1}^{-+} \setminus P_{c_2}^{-+}$, then we must have the order p_1, p_2, q_2, q_1 along $\partial \text{rconv}(S \cup \{p_1, p_2\})$ since $q_2 \in P_{c_2}^{-+}$ by Claim 4.8, a contradiction. In the case of $q_2 \in P_{c_2}^{-+} \setminus P_{c_1}^{-+}$, we also attain a contradiction in the same argument. Therefore, $q_1, q_2 \in P_{c_1}^{-+} \cap P_{c_2}^{-+} = P_{c'}^{-+}$.

Next, suppose that $z \in I_c^{+-}$. In this case, we choose c' so that $P_{c'}^{+-} = P_{c_1}^{+-} \cap P_{c_2}^{+-}$. First, we show that this is possible. Since $z \in I_c^{+-}$, the four points p_1, p_2, c_1, c_2 appear in this order along ∂I_c^{+-} . If $c_1, c_2 \in ch_c^+$, then c_2 lies in between c and c_1 , so we take $c' = c_1$. See Fig. 7 (a). If $c_1, c_2 \in cv_c^-$, then c_1 lies in between c and c_2 , so we take $c' = c_2$. Thus, in the above two cases, c' is well defined as desired.

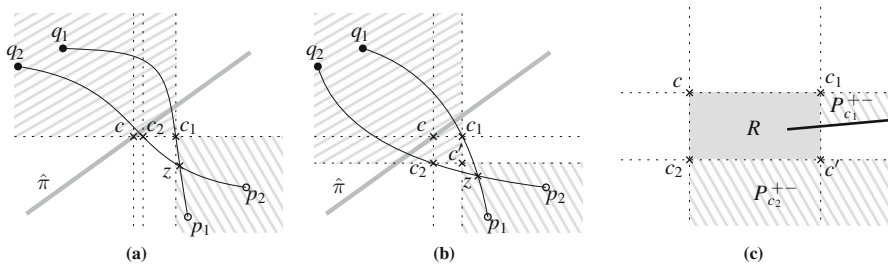


Fig. 7 Illustration to the proof of Claim 4.8 when $z \in I_c^{+-}$. (a) If $c_1, c_2 \in ch_c^+$, we have $c_2 \in cc_1$, and take $c' = c_1$. (b) If $c_1 \in cv_c^-$ and $c_2 \in ch_c^+$, we take c' to be the opposite corner of c in the rectangle R . Shaded regions in (a) and (b) depict $P_{c'}^{+-}$ and $P_{c'}^{+-}$. (c) If c_1c' is not contained in P , we have $P_{c_1}^{+-} \cap P_{c_2}^{+-} = \emptyset$

Otherwise, by the order, we have $c_1 \in ch_c^+$ and $c_2 \in cv_c^-$. In this case, we consider the axis-parallel rectangle R determined by two opposite corners c_1 and c_2 . Then, c' is chosen to be the opposite corner of c . See Fig. 7(b). We show that $R \subset P$, so the choice of c' is feasible. Since P is a simple polygon and $cc_1 \cup cc_2 \subset P$, it suffices to show that $c_1c' \cup c_2c' \subset P$. Suppose to the contrary that c_1c' is not contained in P . Then, the downward half-chord from c_1 is a proper subset of c_1c' , and thus $P_{c_1}^{+-} \cap P_{c_2}^{+-} = \emptyset$. See Fig. 7(c). On the other hand, we have $\pi_2(p_i, c_i) \subset P_{c_i}^{+-}$ by Lemma 3.4 since $p_i \in P_{c_i}^{+-}$ by Claim 4.6. Since $z \in I_c^{+-}$, we indeed have $z \in \pi_2(p_1, c_1) \cap \pi_2(p_2, c_2)$ and thus $z \in P_{c_1}^{+-} \cap P_{c_2}^{+-}$, implying a nonempty intersection between $P_{c_1}^{+-}$ and $P_{c_2}^{+-}$, a contradiction. Thus, we have $c_1c' \subset P$. In the same argument, we can show that $c_2c' \subset P$.

By our choice of c' , observe that

$$P_{c_1}^{+-} \cup P_{c_2}^{+-} \subseteq P_{c'}^{+-} \text{ and } P_{c'}^{+-} = P_{c_1}^{+-} \cap P_{c_2}^{+-}.$$

Since $q_i \in P_{c_i}^{+-}$ by Claim 4.8, this already proves that $q_1, q_2 \in P_{c'}^{+-}$. In order to see that $p_1, p_2 \in P_{c'}^{+-}$, observe that $\partial \text{rconv}(S \cup \{p_1, p_2\})$ intersects the three sets

$$P_{c_2}^{+-} \setminus P_{c_1}^{+-}, \quad P_{c_1}^{+-} \cap P_{c_2}^{+-}, \quad P_{c_1}^{+-} \setminus P_{c_2}^{+-}$$

in this order along $\partial \text{rconv}(S \cup \{p_1, p_2\})$, if nonempty. Thus, if $p_1 \in P_{c_1}^{+-} \setminus P_{c_2}^{+-}$, then we must have the order p_2, p_1, q_1, q_2 along $\partial \text{rconv}(S \cup \{p_1, p_2\})$ since $p_2 \in P_{c_2}^{+-}$ by Claim 4.8, a contradiction. In the case of $p_2 \in P_{c_2}^{+-} \setminus P_{c_1}^{+-}$, we also attain a contradiction in the same argument. Therefore, $p_1, p_2 \in P_{c_1}^{+-} \cap P_{c_2}^{+-} = P_{c'}^{+-}$. $\square$

Now, we are ready to achieve the final contradiction. Let $c' \in P$ be such a point described in Claim 4.9. Then, we have $d(p_i, q_i) = d(p_i, c') + d(c', q_i)$ for $i \in \{1, 2\}$ by Lemma 3.5. Since $f_S(p_i) = q_i$ and $q_i \in P_{c'}^{+-}$, we have $f_S^{+-}(c') = q_i$ for each $i \in \{1, 2\}$ by Lemma 4.3. This leads to a contradiction since $f_S^{+-}(c')$ is uniquely determined by definition.

Consequently, there are no two distinct extreme points $q_1, q_2 \in S$ of S such that $q_1, q_2 \in P^+$, implying that there is at most one extreme point of S in P^+ or P^- . This completes the proof of Theorem 4.1.

5 L_1 Geodesic Center

In this section, we investigate the set $\text{cen}(S)$ of L_1 geodesic centers of S in P . Recall that an L_1 geodesic center c of S minimizes $\Phi_S(c')$ over all $c' \in P$, so $\Phi_S(c) = \text{rad}(S)$. Another consequence from the discussions in the previous section is the following.

Lemma 5.1 *For any nonempty compact subset $S \subseteq P$, there is a diametral pair (χ_1, χ_2) of S such that $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$.*

Proof Fix any center $c \in \text{cen}(S)$. Then, by Lemma 4.4, there exist $\chi_1, \chi_2 \in F_S(c)$ such that (i) $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$ and (ii) $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$. By property (ii), χ_1 and χ_2 are extreme points of S . Then, (χ_1, χ_2) is a diametral pair, that is, $d(\chi_1, \chi_2) = \text{diam}(S)$, since $d(\chi_1, c) + d(c, \chi_2) = 2 \text{rad}(S)$ while $d(q, q') \leq d(q, c) + d(c, q') \leq 2 \text{rad}(S)$ for any $q, q' \in S$. $\square$

The above lemma and its proof indeed show the following.

Corollary 5.2 *For any nonempty compact subset $S \subseteq P$, it holds that $\text{rad}(S) = \text{diam}(S)/2$.*

Bae et al. [3] considered a special case where $S = P$, and proved that $\text{diam}(P) = 2 \text{rad}(P)$ by using a Helly-type theorem: any family of L_1 geodesic balls has Helly number at most two. It is worth noting that we generalize it to any compact subset S of P with a relatively direct argument in terms of extreme points of S .

For $p \in P$ and $r \in \mathbb{R}$, let $B_p(r) := \{x \in P \mid d(x, p) \leq r\}$ be the L_1 geodesic ball at p with radius r . Bae et al. [3] also exhibited several basic properties of the L_1 geodesic balls; among them is that $B_p(r)$ is relatively convex for any $p \in P$ and $r \in \mathbb{R}$.

We fully characterize the set $\text{cen}(S)$ of all centers of S by using those known results.

Lemma 5.3 *For any nonempty compact subset $S \subseteq P$, $\text{cen}(S)$ is equal to the intersection of at most four geodesic balls $\bigcap_{\chi \in X} B_\chi(\text{rad}(S))$, where X is the set of extreme points of S .*

Proof For $p \in P$ and $r \in \mathbb{R}$, let $B_p(r) := \{x \in P \mid d(x, p) \leq r\}$ be the L_1 geodesic ball at p with radius r . Consider the intersection $B(r) := \bigcap_{q \in S} B_q(r)$ for $r \in \mathbb{R}$. If $B(r)$ is nonempty, then for any $p \in B(r)$ we have $d(p, q) \leq r$ for any $q \in S$, that is,

$$d(p, f_S(p)) = \max_{q \in S} d(p, q) \leq r.$$

On the other hand, if $d(p, f_S(p)) \leq r$ for $p \in P$, then $p \in B(r)$. Thus, we have $p \in B(r)$ if and only if $d(p, f_S(p)) \leq r$. This implies that $B(r)$ is equal to the intersection of the four geodesic balls at each of the four extreme points of S by Theorem 4.1. That is, $B(r) = \bigcap_{\chi \in X} B_\chi(r)$. In addition, observe that $B(r) = \emptyset$ for $r < \text{rad}(S)$. Thus, it holds that

$$\text{cen}(S) = B(\text{rad}(S)) = \bigcap_{\chi \in X} B_\chi(\text{rad}(S)),$$

as claimed. $\square$

In the following, we assume that S consists of at least two points. Then, the set X of extreme points of S consists of at least two and at most four points. Let (χ_1, χ_2) be a diametral pair such that $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$. Such a diametral pair exists by Lemma 5.1. Since $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$, both χ_1 and χ_2 are extreme points of S , that is, $\chi_1, \chi_2 \in X$. By Corollary 5.2, we know that $\text{diam}(S) = 2 \text{rad}(S)$. Thus, $B_{\chi_1}(\text{rad}(S))$ and $B_{\chi_2}(\text{rad}(S))$ intersect only in their boundaries. Let $B := B_{\chi_1}(\text{rad}(S)) \cap B_{\chi_2}(\text{rad}(S))$. Since any L_1 geodesic ball is relatively convex, as shown in [3], we observe that $B \cap \partial P$ is either an empty set, a single point, or two points. Again by the relative convexity, B already forms a line segment of slope 1 or -1 , since the boundary of any L_1 geodesic ball in the interior of P consists of line segments of slope 1 or -1 . By Lemma 5.3, it holds that $\text{cen}(S) \subseteq B$. This implies the following.

Corollary 5.4 *The set $\text{cen}(S)$ of centers of any nonempty compact subset $S \subseteq P$ forms a line segment of slope 1 or -1 , unless it consists of a single point.*

Let $c_1, c_2 \in P$ be the endpoints of the segment $\text{cen}(S)$. By Lemma 5.3, we know that $\text{cen}(S) = B \cap \bigcup_{\chi \in X \setminus \{\chi_1, \chi_2\}} B_\chi(\text{rad}(S))$. Thus, if $|X| \geq 3$, then a third extreme $\chi_3 \in X$ with $\chi_3 \neq \chi_1, \chi_2$ determines an endpoint c_1 or c_2 of $\text{cen}(S)$ as the intersection $B \cap \partial B_{\chi_3}(\text{rad}(S))$. More precisely, we observe the following.

Corollary 5.5 *Suppose that $X = \{\chi_1, \chi_2, \dots, \chi_k\}$ with $3 \leq k \leq 4$, and (χ_1, χ_2) is a diametral pair of S with $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$. Then, for each $3 \leq i \leq k$, $\partial B_{\chi_1}(\text{rad}(S)) \cap \partial B_{\chi_2}(\text{rad}(S)) \cap \partial B_{\chi_i}(\text{rad}(S))$ determines an endpoint of the segment $\text{cen}(S)$.*

6 L_1 Geodesic Farthest-Neighbor Voronoi Diagram

We then turn our attention to the L_1 geodesic farthest-neighbor Voronoi diagram. Given a set S of sites in P , its L_1 geodesic farthest-neighbor Voronoi diagram $\text{FVD}(S)$ is a partition of P into regions according to the farthest-neighbor relation between P and S . A common degenerate case of Voronoi diagrams occurs when a point $p \in P$ has four or more equidistant sites in S . There are two popular approaches in the literature to resolve such a degenerate case: assume a general position or impose a total order $<$ on S . We take the latter as done so far to give a precise definition of $\text{FVD}(S)$.

The L_1 geodesic farthest-neighbor Voronoi region $FR(q, S)$ for each $q \in S$ is defined to be

$$FR(q, S) := \{p \in P \mid f_S(p) = q\}.$$

Then, the L_1 geodesic farthest-neighbor Voronoi diagram $\text{FVD}(S)$ is defined to be

$$\text{FVD}(S) := \bigcup_{q \in S} \partial FR(q, S) \setminus \partial P,$$

the union of the boundaries of each farthest-neighbor Voronoi region, except ∂P .

By definition, the Voronoi region $FR(q, S)$ for $q \in S$ is nonempty if and only if q is an extreme point of S . By Theorem 4.1, this implies that $FVD(S)$ coincides with the diagram of at most four points in S . This enables us to define the Voronoi diagram $FVD(S)$ for any nonempty compact subset $S \subseteq P$, even if S consists of an infinite number of points.

Let $X \subseteq S$ be the set of extreme points of S . Lemma 5.1 guarantees the existence of a diametral pair (χ_1, χ_2) of S with $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$, so $\chi_1, \chi_2 \in X$. We first observe the following property of such a diametral pair.

Lemma 6.1 *Let (χ_1, χ_2) be any diametral pair of S with $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$. Then, there exist $\sigma_1, \sigma_2 \in \{+, -\}$ such that $\chi_1 = f_S^{\sigma_1 \sigma_2}(c)$ and $\chi_2 = f_S^{\sigma_1 \sigma_2}(c)$ for all $c \in \text{cen}(S)$. Moreover, if $\text{cen}(S)$ forms a line segment of positive length, then $\sigma_1 = \sigma_2$ when $\text{cen}(S)$ is of slope -1 , or $\sigma_1 = \overline{\sigma_2}$, otherwise.*

Proof By Corollary 5.2, we have $d(\chi_1, \chi_2) = 2 \text{rad}(S)$. Let $c \in \text{cen}(S)$ be any center of S . Also, by Lemma 5.3, we have $d(\chi_1, c) = d(\chi_2, c) = \text{rad}(S)$, and thus $d(\chi_1, \chi_2) = d(\chi_1, c) + d(c, \chi_2)$. Hence, Lemma 3.5 implies that $\chi_1 \in P_c^{\sigma_1 \sigma_2}$ and $\chi_2 \in P_c^{\sigma_1 \sigma_2}$ for some $\sigma_1, \sigma_2 \in \{+, -\}$. This also implies that c lies on a shortest path $\pi \in \Pi(\chi_1, \chi_2)$ from χ_1 to χ_2 . Since $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$, Lemma 4.3 implies that $\chi_1 = f_S^{\sigma_1 \sigma_2}(c)$ and $\chi_2 = f_S^{\sigma_1 \sigma_2}(c)$. Since c can be arbitrarily chosen from $\text{cen}(S)$, we prove the first part of the lemma.

In order to see the second part, assume that $\sigma_1 = \sigma_2 = -$. Recall that $\text{cen}(S)$ forms a line segment of slope 1 or -1 , unless it is degenerate to a point by Corollary 5.4. Suppose that $\text{cen}(S)$ is a line segment of positive length and slope 1. Let $c \in \text{cen}(S)$ be any center of S that is not an endpoint of $\text{cen}(S)$. Then, both I_c^{--} and I_c^{++} contains a center $c' \in \text{cen}(S)$ of S . Pick any $c' \in \text{cen}(S) \cap I_c^{--}$. Then, by Lemma 3.5, we have $d(\chi_2, c') = d(\chi_2, c) + d(c, c') > \text{rad}(S)$, a contradiction, since for any $c' \in \text{cen}(S)$ and any $q \in S$, it holds that $d(c', q) \leq \text{rad}(S)$.

Therefore, if $\sigma_1 = \sigma_2$, then $\text{cen}(S)$ is a line segment of slope -1 or a point. In the same argument, we can show that if $\sigma_1 \neq \sigma_2$, then $\text{cen}(S)$ is a line segment of slope 1 or a point. Conversely, if $\text{cen}(S)$ is not a point but is a segment of slope 1, then we have $\sigma_1 = \sigma_2$; if $\text{cen}(S)$ is not a point but is a segment of slope -1 , then we have $\sigma_1 \neq \sigma_2$. This proves the second part of the lemma. $\square$

Note that if there are two distinct such pairs (χ_1, χ_2) and (χ'_1, χ'_2) , then Lemma 6.1 implies that the four points $\chi_1, \chi_2, \chi'_1, \chi'_2$ must be all distinct. Since $|X| \leq 4$ by Theorem 4.1, this implies that there are at most two such pairs.

We then observe the following.

Lemma 6.2 *Suppose that $\text{cen}(S)$ forms a line segment of slope $(\sigma 1)$ for $\sigma \in \{+, -\}$ or a point. Let c^- and c^+ be the left and right endpoints of $\text{cen}(S)$. Then, the following hold:*

- $I_c^{-\sigma} \subseteq FR(f_S^{+\sigma}(c), S)$ and $I_c^{+\sigma} \subseteq FR(f_S^{-\sigma}(c), S)$ for any $c \in \text{cen}(S)$.
- $I_{c^-}^{-\sigma} \subseteq FR(f_S^{+\sigma}(c^-), S)$ and $I_{c^+}^{+\sigma} \subseteq FR(f_S^{-\sigma}(c^+), S)$.
- $H_{c^-}^- \setminus \{c^-\} \subseteq FR(f_S^{+\sigma}(c^-), S)$ and $V_{c^-}^{\overline{\sigma}} \setminus \{c^-\} \subseteq FR(f_S^{\sigma 1 \sigma}(c^-), S)$ where $\sigma_1, \sigma_2 \in \{+, -\}$.

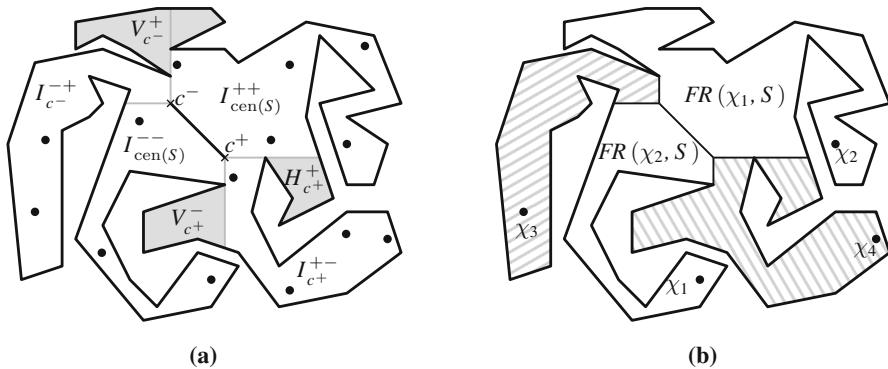


Fig. 8 (a) The partition around the center $\text{cen}(S)$. Points in S are depicted as dots. (b) Illustration to $\text{FVD}(S)$. Shaded regions depict $\text{FR}(\chi_3, S)$ and $\text{FR}(\chi_4, S)$, respectively. In this example, S has four extreme points $\chi_1, \chi_2, \chi_3, \chi_4$ and we assume that $\chi_3 < \chi_1 < \chi_4 < \chi_2$

- $H_{c^+}^+ \setminus \{c^+\} \subseteq \text{FR}(f_S^{-\sigma'_2}(c^+), S)$ and $V_{c^+}^+ \setminus \{c^+\} \subseteq \text{FR}(f_S^{\sigma'_1 \bar{\sigma}}(c^+), S)$ where $\sigma'_1, \sigma'_2 \in \{+, -\}$.

Proof Let (χ_1, χ_2) be a diametral pair with $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$. Such a pair exists by Lemma 5.1. Assume without loss of generality that $\sigma = -$, so $\text{cen}(S)$ forms a line segment of slope -1 or a point. Then, the lemma states the following:

- $I_c^{--} \subseteq \text{FR}(f_S^{++}(c), S)$ and $I_c^{++} \subseteq \text{FR}(f_S^{--}(c), S)$ for any $c \in \text{cen}(S)$.
- $I_c^{-+} \subseteq \text{FR}(f_S^{+-}(c^-), S)$ and $I_c^{+-} \subseteq \text{FR}(f_S^{-+}(c^+), S)$.
- $H_c^- \setminus \{c^-\} \subseteq \text{FR}(f_S^{+-}(c^-), S)$ or $H_c^- \setminus \{c^-\} \subseteq \text{FR}(f_S^{++}(c^-), S)$, and $V_c^+ \setminus \{c^-\} \subseteq \text{FR}(f_S^{+-}(c^-), S)$ or $V_c^+ \setminus \{c^-\} \subseteq \text{FR}(f_S^{--}(c^-), S)$.
- $H_c^+ \setminus \{c^+\} \subseteq \text{FR}(f_S^{-+}(c^+), S)$ or $H_c^+ \setminus \{c^+\} \subseteq \text{FR}(f_S^{--}(c^+), S)$, and $V_c^- \setminus \{c^+\} \subseteq \text{FR}(f_S^{-+}(c^+), S)$ or $V_c^- \setminus \{c^+\} \subseteq \text{FR}(f_S^{++}(c^+), S)$.

If $\text{cen}(S)$ is a line segment of positive length, then, by Lemma 6.1, we have $\{\chi_1, \chi_2\} = \{f_S^{--}(c), f_S^{++}(c)\}$. If $\text{cen}(S)$ is a point, then we have either $\{\chi_1, \chi_2\} = \{f_S^{--}(c), f_S^{++}(c)\}$ or $\{\chi_1, \chi_2\} = \{f_S^{-+}(c), f_S^{+-}(c)\}$. Thus, without loss of generality, we assume that $\chi_1 = f_S^{--}(c)$ and $\chi_2 = f_S^{++}(c)$ for all $c \in \text{cen}(S)$, in either case. In the following, the case where $\text{cen}(S)$ is a point is considered as a special case such that $c^+ = c^-$.

Since $\chi_1 = f_S^{--}(c)$ and $\chi_2 = f_S^{++}(c)$ for all $c \in \text{cen}(S)$, Lemma 4.2 implies that for any $p \in I_c^{--}$ with some $c \in \text{cen}(S)$, we have $f_S(p) = \chi_2 = f_S^{++}(c)$, while for any $p \in I_c^{++}$ with some $c \in \text{cen}(S)$, we have $f_S(p) = \chi_1 = f_S^{--}(c)$. Hence, $I_c^{--} \subseteq \text{FR}(f_S^{++}(c), S)$ and $I_c^{++} \subseteq \text{FR}(f_S^{--}(c), S)$ for all $c \in \text{cen}(S)$.

Now, consider the left endpoint c^- of $\text{cen}(S)$. We first prove that $I_{c^-}^{-+} \subseteq \text{FR}(f_S^{+-}(c^-), S)$. If $I_{c^-}^{-+} = \emptyset$, then it is trivially true. Otherwise, for any $p \in I_{c^-}^{-+}$, we have $f_S(p) = f_S^{+-}(c^-)$ by Lemma 4.3, so $I_{c^-}^{-+} \subseteq \text{FR}(f_S^{+-}(c^-), S)$.

We then show that $H_{c^-}^- \setminus \{c^-\} \subseteq \text{FR}(f_S^{+-}(c^-), S)$ or $H_{c^-}^- \setminus \{c^-\} \subseteq \text{FR}(f_S^{++}(c^-), S)$. Note that $f_S^{++}(c^-) = \chi_2$, and $\chi_2 \in F_S(c^-)$ since $d(\chi_2, c^-) = \text{rad}(S)$. Also, note that $H_{c^-}^- = P_{c^-}^{--} \cap P_{c^-}^{-+}$.

There are two cases: whether $f_S^{+-}(c^-) \in F_S(c^-)$ or not. Suppose the former case that $f_S^{+-}(c^-) \in F_S(c^-)$. Since $\chi_2 \in P_{c^-}^{++}$, $f_S^{+-}(c^-) \in P_{c^-}^{+-}$, and both are farthest neighbors of c^- , Lemma 4.2 implies that for any $p \in H_{c^-}^-$, we have $\chi_2, f_S^{+-}(c^-) \in F_S(p)$. On the other hand, for any $p \in H_{c^-}^- \setminus \{c^-\}$ and any $q \in S \setminus (P_{c^-}^{++} \cup P_{c^-}^{+-})$, we have a strict inequality that

$$d(p, q) < d(p, c^-) + d(c^-, q) \leq d(p, c^-) + d(c^-, \chi_2) = d(p, \chi_2) = \Phi_S(p),$$

by Lemma 3.5. This implies that no $q \in S \setminus (P_{c^-}^{++} \cup P_{c^-}^{+-})$ is a farthest neighbor of any $p \in H_{c^-}^- \setminus \{c^-\}$. Thus, $f_S(p) \in \{\chi_2, f_S^{+-}(c^-)\}$ for any $p \in H_{c^-}^- \setminus \{c^-\}$. Moreover, if $f_S^{+-}(c^-) < \chi_2$, then we have $f_S(p) = f_S^{+-}(c^-)$ for all $p \in H_{c^-}^- \setminus \{c^-\}$; otherwise, if $\chi_2 < f_S^{+-}(c^-)$, then we have $f_S(p) = \chi_2$ for all $p \in H_{c^-}^- \setminus \{c^-\}$, since $f_S^{+-}(c^-), \chi_2 \in F_S(p)$ for all $p \in H_{c^-}^-$. This implies that $H_{c^-}^- \setminus \{c^-\} \subseteq FR(f_S^{+-}(c^-), S)$ or $H_{c^-}^- \setminus \{c^-\} \subseteq FR(\chi_2, S)$.

In the latter case where $f_S^{+-}(c^-) \notin F_S(c^-)$, the above argument implies that no $q \in S \setminus P_{c^-}^{++}$ is a farthest neighbor of any $p \in H_{c^-}^- \setminus \{c^-\}$, since $d(f_S^{+-}(c^-), c^-) < d(\chi_2, c^-) = \text{rad}(S)$. Thus, in this case, we have $H_{c^-}^- \setminus \{c^-\} \subseteq FR(\chi_2, S)$.

In the similar way, we can show that $V_{c^-}^+ \setminus \{c^-\} \subseteq FR(f_S^{+-}(c^-), S)$ or $V_{c^-}^+ \setminus \{c^-\} \subseteq FR(\chi_2, S)$.

Furthermore, an analogous argument shows that $I_{c^+}^{+-} \subseteq FR(f_S^{+-}(c^+), S)$, $H_{c^+}^+ \setminus \{c^+\} \subseteq FR(f_S^{+-}(c^+), S)$ or $H_{c^+}^+ \setminus \{c^+\} \subseteq FR(f_S^{+-}(c^+), S)$, and $V_{c^+}^+ \setminus \{c^+\} \subseteq FR(f_S^{+-}(c^+), S)$ or $V_{c^+}^+ \setminus \{c^+\} \subseteq FR(f_S^{+-}(c^+), S)$. This completes our proof of the lemma. $\square$

Lemma 6.2 fully describes the farthest-neighbor Voronoi diagram $\text{FVD}(S)$, according to the shape of $\text{cen}(S)$. Assume without loss of generality that $\text{cen}(S)$ forms a line segment of slope -1 or a point. Let $I_{\text{cen}(S)}^{--} := \bigcup_{c \in \text{cen}(S)} I_c^{--}$ and $I_{\text{cen}(S)}^{++} := \bigcup_{c \in \text{cen}(S)} I_c^{++}$. Then, observe that the eight subsets $I_{\text{cen}(S)}^{--}, V_{c^-}^+, I_{c^+}^{+-}, H_{c^+}^+, I_{\text{cen}(S)}^{++}, V_{c^-}^+, I_{c^-}^{+-}$, and $H_{c^-}^-$ form a partition of P around $\text{cen}(S)$. See Fig. 8 for an illustration. Lemma 6.2 describes to which Voronoi region each of these eight subsets of P belongs. Note that each of the four subsets $I_{\text{cen}(S)}^{--}, I_{c^+}^{+-}, I_{\text{cen}(S)}^{++}$, and $I_{c^-}^{+-}$ may be empty, when c^- or c^+ lies on ∂P . If $\text{cen}(S) = \{c\}$ consists of a single point, then we have $c^+ = c^- = c$, $I_{\text{cen}(S)}^{--} = I_c^{--}$, and $I_{\text{cen}(S)}^{++} = I_c^{++}$. Otherwise, $\text{cen}(S)$ forms a line segment of positive length. Then, since $\text{cen}(S)$ is of slope -1 , by Lemma 6.1, we have $\chi_1, \chi_2 \in X$ such that $\chi_1 = f_S^{--}(c)$ and $\chi_2 = f_S^{++}(c)$ for any $c \in \text{cen}(S)$. Lemma 6.2 tells us that $I_{\text{cen}(S)}^{++} \subseteq FR(\chi_1, S)$ and $I_{\text{cen}(S)}^{--} \subseteq FR(\chi_2, S)$. On the other hand, if $I_{c^{\sigma'}^+}^{\sigma'} \neq \emptyset$ for any $\sigma' \in \{+, -\}$, then the endpoint $c^{\sigma'}$ is not a boundary point in ∂P . In particular, if $f_S^{\sigma'\sigma'}(c^{\sigma'}) \notin \{\chi_1, \chi_2\}$, then we have a third extreme point $\chi_3 = f_S^{\sigma'\sigma'}(c^{\sigma'})$ as described in Corollary 5.5.

Since the boundaries of any two of the eight subsets around $\text{cen}(S)$ always intersect in a subset of a half-chord from c^- or from c^+ , we conclude the following.

Theorem 6.3 *For any compact subset $S \subseteq P$ with $|S| \geq 2$, its L_1 geodesic farthest-neighbor Voronoi diagram $\text{FVD}(S)$ consists of $\text{cen}(S)$ and a subset of the following four segments: two segments that are subsets of half-chords from each endpoint of $\text{cen}(S)$.*

7 Algorithms

Now, we are ready to describe our algorithms that compute the extreme points X of S , the diameter, radius, center of S and the farthest-neighbor Voronoi diagram $\text{FVD}(S)$. We keep the generality by setting S to be any nonempty compact subset of P , while an operation that computes $f_S(p)$ for any $p \in P$ is supposed to be processed in time at most T as a black box.

We first describe how to compute the set X of extreme points of S . Pick any $q_0 \in S$. Let $q_i := f_S(q_{i-1})$ for $i \geq 1$, and compute q_i until we have $q_{k+1} = q_{k-1}$ for some $k \geq 2$. By Theorem 4.1, this ends up with $k \leq 4$. If $k = 4$, then let $\chi_i := q_i$ for each $i \in \{1, 2, 3, 4\}$, and we are done as $X = \{\chi_1, \chi_2, \chi_3, \chi_4\}$ by Theorem 4.1. Otherwise, we let $\chi_1 := q_{k-1}$ and $\chi_2 := q_k$. Note that $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$. Let $r := d(\chi_1, \chi_2)/2$. Then, we compute $B_{\chi_1}(r)$ and $B_{\chi_2}(r)$, and their intersection $B_{\chi_1}(r) \cap B_{\chi_2}(r)$. Since $r = d(\chi_1, \chi_2)/2$, $B_{\chi_1}(r) \cap B_{\chi_2}(r)$ forms a line segment z^-z^+ of slope 1 or -1 , where z^- is to the left of z^+ , possibly being a point $z^- = z^+$. Without loss of generality, assume that z^-z^+ is of slope -1 . For each $\sigma \in \{+, -\}$, let $p^\sigma \in P$ be any point in $I_{z^\sigma}^{\sigma\bar{\sigma}}$ if $I_{z^\sigma}^{\sigma\bar{\sigma}}$ is nonempty, or let $p^\sigma := z^\sigma$, otherwise, if $I_{z^\sigma}^{\sigma\bar{\sigma}} = \emptyset$. Let $\chi_3 := f_S(p^-)$ and $\chi_4 := f_S(p^+)$. Then, we have $X = \{\chi_1, \chi_2, \chi_3, \chi_4\}$.

Lemma 7.1 *Let $S \subseteq P$ be a given compact subset, and suppose that $f_S(p)$ for any $p \in P$ can be computed in T time. The above algorithm correctly computes the set X of extreme points of S in $O(n + T)$ time.*

Proof First, we prove the correctness of our algorithm. If $k = 4$, then the four points $q_1, q_2, q_3, q_4 \in S$ are all distinct. Moreover, for each $i \in \{1, 2, 3, 4\}$, we have $q_i = f_S(q_{i-1})$, that is, q_i is an extreme point of S . Since there are at most four extreme points of S by Theorem 4.1, we have $X = \{q_1, q_2, q_3, q_4\}$.

Otherwise, if $k < 4$, then by our algorithm, we have $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$. There are two cases: whether (χ_1, χ_2) is a diametral pair or not. First, suppose the former case. Then, we have $r = d(\chi_1, \chi_2)/2 = \text{rad}(S)$ by Corollary 5.2, and $\text{cen}(S) \subseteq z^-z^+$ by Lemma 5.3. Without loss of generality, assume that z^-z^+ is a line segment of slope -1 . Then, $\text{cen}(S)$ also forms a line segment c^-c^+ of slope -1 , where c^- and c^+ denote the left and right endpoints of $\text{cen}(S)$. We let $\chi_3 = f_S(p^-)$ and $\chi_4 = f_S(p^+)$ by our algorithm. By our choice of p^- and p^+ , either $p^- \in I_{c^-}^{-+}$ or $p^- = c^-$. If $p^- \in I_{c^-}^{-+}$, then the endpoint c^- of $\text{cen}(S)$ is either equal to z^- or an intersection point between z^-z^+ and $\partial B_\chi(\text{rad}(S))$ for a third extreme point χ of S other than χ_1 and χ_2 by Corollary 5.5. If $c^- = z^-$, then we have $f_S(p^-) \in \{\chi_1, \chi_2\}$, while, otherwise, we have $\chi_3 = f_S(p^-) = \chi$. Otherwise, if $p^- = c^-$, then $I_{c^-}^{-+} = \emptyset$ and thus $c^- \in \partial P$. In this case, either $\chi_3 = f_S(c^-) \in \{\chi_1, \chi_2\}$ or a third extreme point χ . Hence, in either case, we have $\chi_3 \in \{\chi_1, \chi_2\}$ or we find a third extreme point χ_3 of S that determines the endpoint c^- of $\text{cen}(S)$. Similarly, we have $\chi_4 \in \{\chi_1, \chi_2\}$ or χ_4

is another extreme point of S that determines the endpoint c^+ of $\text{cen}(S)$. This implies that $X = \{\chi_1, \chi_2, \chi_3, \chi_4\}$.

Now, suppose the latter case where (χ_1, χ_2) is not a diametral pair. Then, the diameter $\text{diam}(S)$ is determined by the other two extreme points of S . In this case, by Lemma 5.3, z^-z^+ must cross $\text{cen}(S)$ at the midpoint of $\text{cen}(S)$. Also, $\text{cen}(S)$ forms a line segment c^-c^+ of slope 1, where c^- and c^+ denote the left and right endpoints of $\text{cen}(S)$. Thus, by our choice of p^- and p^+ , it holds that $p^- \in I_c^{-+}$ and $p^+ \in I_c^{+-}$, where c denotes the midpoint of $\text{cen}(S)$. Then, $\chi_3 = f_S(p^-)$ and $\chi_4 = f_S(p^+)$ form a diametral pair of S such that $f_S(\chi_3) = \chi_4$ and $f_S(\chi_4) = \chi_3$ by Lemmas 5.1 and 6.1. Consequently, we have $X = \{\chi_1, \chi_2, \chi_3, \chi_4\}$. This completes the proof of correctness.

For the time complexity of our algorithm, observe that the algorithm performs a constant number of applications of $f_S(\cdot)$, and computes two L_1 geodesic balls and their intersection. Computing a geodesic ball $B_p(r)$ can be done in $O(n)$ time by building the shortest path map and traversing each cell of the map [7,8]. Hence, our algorithm is done in $O(n + T)$ time. $\square$

The diameter, radius, center, and farthest-neighbor Voronoi diagram of S can be computed in the same time bound.

Lemma 7.2 *Let $S \subseteq P$ be a given compact subset, and suppose that the set of extreme points of S is known. Then, the following can be computed in $O(n)$ time: $\text{diam}(S)$, $\text{rad}(S)$, $\text{cen}(S)$, and $\text{FVD}(S)$.*

Proof Let X be the set of extreme points of S . Note that $\text{diam}(S) = \max_{\chi, \chi' \in X} d(\chi, \chi')$. Thus, $\text{diam}(S)$ and a diametral pair can be computed in additional $O(n)$ time [8], as $|X| \leq 4$ by Theorem 4.1. By Corollary 5.2, we have $\text{rad}(S) = \text{diam}(S)/2$. The set $\text{cen}(S)$ can be computed by intersecting at most four geodesic balls $B_\chi(\text{rad}(S))$ for $\chi \in X$ by Lemma 5.3. This can be done in additional $O(n)$ time by computing the shortest path maps [8]. After computing $\text{cen}(S)$, the farthest-neighbor Voronoi diagram $\text{FVD}(S)$ can be found by considering the eight subsets around $\text{cen}(S)$ by Lemma 6.2. As $\text{FVD}(S)$ consists of at most five segments, it can be found in additional $O(n)$ time. $\square$

Now, we describe the subprocedure that computes $f_S(p)$ for any $p \in P$. Here, we assume that S is a finite set of m points.

Lemma 7.3 *Let S be a set of m points in P . Then, $f_S(p)$ for any $p \in P$ can be computed in $O(n + m \log n)$ time. If the order of $S \cap \partial \text{rconv}(S)$ along $\partial \text{rconv}(S)$ is provided, then $O(n + m)$ time is sufficient.*

Proof As a preprocessing, we build in $O(n)$ time the data structure of Guibas and Hershberger [7] that evaluates $d(p, q)$ for any $p, q \in P$ in $O(\log n)$ time. Given any $p \in P$, we compute $d(p, q)$ for all $q \in S$, and gather the set $F_S(p)$ of farthest neighbors of p in $O(m \log n)$ time. And pick the least element in $F_S(p)$ with respect to the total order $<$, and report it as $f_S(p)$. This takes $O(n + m \log n)$ time.

If the order of $S \cap \partial \text{rconv}(S)$ along $\partial \text{rconv}(S)$ is known, then we can apply the fast matrix search technique of Hershberger and Suri [10]. This takes $O(n + m)$ time. $\square$

Another interesting case is when $S = P$. Since $\text{rconv}(P) = P$, in this case, we know the order of points $P \cap \partial \text{rconv}(P) = \partial P$. Moreover, since $f_P(p)$ is always a vertex of P , we have the following corollary.

Corollary 7.4 *For any $p \in P$, $f_P(p)$ can be computed in $O(n)$ time.*

Combining all these results, we obtain the following theorems.

Theorem 7.5 *Let P be a simple n -gon and S be a set of m points in P . Then, the set of L_1 geodesic extreme points of S , $\text{diam}(S)$, $\text{rad}(S)$, $\text{cen}(S)$, and $\text{FVD}(S)$ can be computed in $O(n + m \log n)$ time. If the order of $S \cap \partial \text{rconv}(S)$ along $\partial \text{rconv}(S)$ is provided, then $O(n + m)$ time is sufficient.*

Theorem 7.6 *Let P be a simple n -gon. Then, the set of L_1 geodesic extreme points of P , $\text{diam}(P)$, $\text{rad}(P)$, $\text{cen}(P)$, and $\text{FVD}(P)$ can be computed in $O(n)$ time.*

8 L_1 Geodesic Two-Center

In this section, we address the two-center problem for any compact subset $S \subseteq P$ under the L_1 geodesic distance. The L_1 geodesic two-center problem asks a pair of points $c_1, c_2 \in P$ that minimize $\max_{q \in S} \min\{d(q, c_1), d(q, c_2)\}$. Such a pair (c_1, c_2) is called an L_1 geodesic two-center of S in P , or shortly a two-center of S . Let $\text{rad}_2(S) := \max_{q \in S} \min\{d(q, c_1), d(q, c_2)\}$ be the optimal objective value for the problem, called the two-radius or 2-radius of S . A two-center (c_1, c_2) induces a bipartition (S_1, S_2) of S such that $S_1 = S \cap B_{c_1}(\text{rad}_2(S))$ and $S_2 = S \setminus S_1$. Conversely, a bipartition (S_1, S_2) of S is called optimal if $\text{rad}_2(S) = \max\{\text{rad}(S_1), \text{rad}(S_2)\}$. Note that in general we have $\max\{\text{rad}(S_1), \text{rad}(S_2)\} \geq \text{rad}_2(S)$ if $S_1 \cup S_2 = S$. Given an optimal bipartition (S_1, S_2) of S , observe that any $c_1 \in \text{cen}(S_1)$ and $c_2 \in \text{cen}(S_2)$ form a two-center (c_1, c_2) of S . Thus, the two-center problem is equivalent to finding an optimal bipartition of S .

Another closely related problem is the minmax-diameter bipartition problem that asks a bipartition (S_1, S_2) of S such that $\max\{\text{diam}(S_1), \text{diam}(S_2)\}$ is minimized. Thus, this problem is to compute the 2-diameter $\text{diam}_2(S)$ of S defined to be the minimum value of $\max\{\text{diam}(S_1), \text{diam}(S_2)\}$ over all possible bipartitions (S_1, S_2) of S . In the L_1 geodesic case, the two-center problem is equivalent to the minmax-diameter bipartition problem.

Lemma 8.1 *For any compact subset $S \subseteq P$, it holds that $\text{rad}_2(S) = \text{diam}_2(S)/2$.*

Proof By Corollary 5.2, for any bipartition (S_1, S_2) of S , we have

$$\max\{\text{rad}(S_1), \text{rad}(S_2)\} = \max\{\text{diam}(S_1)/2, \text{diam}(S_2)/2\}.$$

Thus, the lemma follows. $\square$

Thus, if (S_1, S_2) is the optimal solution to the minmax-diameter bipartition problem, then it is also an optimal bipartition for the two-center problem, and vice versa.

In the following, we let X be the set of extreme points of S .

Lemma 8.2 *There exists an optimal bipartition (S_1, S_2) of S such that for each $\chi \in X$, $\chi \in S_1$ if and only if $f_S(\chi) \in S_2$.*

Proof Recall that for any $q \in S$, we have $f_S(q) \in X$. Let (S_1, S_2) be an optimal bipartition of S , and $X_i := X \cap S_i$ for each $i = 1, 2$. Let $a = a(S_1, S_2)$ be the number of pairs $(\chi, f_S(\chi))$ such that $\chi, f_S(\chi) \in X_i$ for some $i \in \{1, 2\}$. Since $|X| \leq 4$ by Theorem 4.1, we have $a \leq 4$. If $a = 0$, then we are done. Suppose that $a \geq 1$. Then, we describe a process that produces another optimal bipartition (S'_1, S'_2) of S such that $a(S'_1, S'_2) < a$, so repeatedly applying the process finally yields an optimal bipartition with the desired property.

Without loss of generality, we assume that there exist $\chi, \chi' \in X_1$ such that $\chi' = f_S(\chi)$. Note that $\text{diam}_2(S) = \max\{\text{diam}(S_1), \text{diam}(S_2)\} \geq d(\chi, \chi') = \Phi_S(\chi)$, and $\Phi_{S_1}(\chi) = d(\chi, \chi') = \Phi_S(\chi) \geq d(\chi, q)$ for any $q \in S$. We distinguish three cases as follows:

- Suppose that $f_S(\chi') \in X_2$ and there is no $\chi'' \in X_2$ with $f_S(\chi'') = \chi$. We let $S'_1 := S_1 \setminus \{\chi\}$ and $S'_2 := S_2 \cup \{\chi\}$. Clearly, we have $\text{diam}(S'_1) \leq \text{diam}(S_1)$. We also have $\text{diam}(S'_2) = \max\{\text{diam}(S_2), \max_{q \in S_2} d(\chi, q)\}$. On the other hand, $\max_{q \in S_2} d(\chi, q) \leq \Phi_S(\chi) = d(\chi, \chi') \leq \text{diam}_2(S)$. Thus, we have $\max\{\text{diam}(S'_1), \text{diam}(S'_2)\} \leq \text{diam}_2(S)$, so (S'_1, S'_2) is another optimal bipartition of S with $a(S'_1, S'_2) = a - 1$.
- Suppose that $f_S(\chi') \in X_1$ and there is no $\chi'' \in X_2$ with $f_S(\chi'') = \chi$. In this case, we let $S'_1 := S_1 \setminus \{\chi'\}$ and $S'_2 := S_2 \cup \{\chi'\}$. As above, $\text{diam}(S'_1) \leq \text{diam}(S_1)$. We have $\text{diam}(S'_2) = \max\{\text{diam}(S_2), \max_{q \in S_2} d(\chi', q)\}$. On the other hand, we have $\max_{q \in S_2} d(\chi', q) \leq \Phi_S(\chi') = d(\chi', f_S(\chi')) \leq \text{diam}(S_1)$ as $\chi', f_S(\chi') \in X_1 \subseteq S_1$. We thus have $\text{diam}(S'_2) \leq \max\{\text{diam}(S_2), \text{diam}(S_1)\} = \text{diam}_2(S)$. Hence, (S'_1, S'_2) is another optimal bipartition of S with $a(S'_1, S'_2) = a - 2$.
- Suppose the last case where there is $\chi'' \in X_2$ such that $f_S(\chi'') = \chi$. In this case, we first let $S''_1 := S_1 \cup \{\chi''\}$ and $S''_2 := S_2 \setminus \{\chi''\}$. Since $\chi = f_S(\chi'')$ and $\chi' = f_S(\chi)$, we have $d(\chi'', \chi) \leq d(\chi, \chi')$. As $\text{diam}(S_1) \geq d(\chi, \chi')$, we have $\text{diam}(S''_1) \leq \text{diam}(S_1)$. Thus, (S''_1, S''_2) is another optimal bipartition of S , while $a(S''_1, S''_2) = a + 1$ as $\chi, \chi', \chi'' \in S''_1$.

On the other hand, there must exist a diametral pair (χ_1, χ_2) in X such that $f_S(\chi_1) = \chi_2$ and $f_S(\chi_2) = \chi_1$ by Lemma 5.1. Since $f_S(\chi'') = \chi$ and $f_S(\chi) = \chi'$, there must exist the fourth extreme point $\chi''' \in X$ such that $\{\chi_1, \chi_2\} = \{\chi', \chi'''\}$. Then, observe that the bipartition (S''_1, S''_2) falls into the second case handled above: $\chi'', \chi \in S''_1$, $f_S(\chi) = \chi' \in S''_2$, and there is no $q \in S''_2 \cap X$ such that $f_S(q) = \chi''$. So, we apply the process for the second case to (S''_1, S''_2) to result in another optimal bipartition (S'_1, S'_2) with $a(S'_1, S'_2) = a(S''_1, S''_2) - 2 = a - 1$.

This completes the proof. $\square$

The following is our key lemma.

Lemma 8.3 *There exists an optimal bipartition (S_1^*, S_2^*) of S such that*

$$S_1^* = S \cap \bigcup_{\chi \in X \cap S_2^*} FR(\chi, S) \quad \text{and} \quad S_2^* = S \cap \bigcup_{\chi \in X \cap S_1^*} FR(\chi, S).$$

Proof We pick any optimal bipartition (S_1, S_2) of S with the property described in Lemma 8.2, that is, there is no $\chi \in X$ such that $\chi, f_S(\chi) \in S_i$ for some $i \in \{1, 2\}$.

Let $X_1 := X \cap S_1$ and $X_2 := X \cap S_2$. We consider a partition of P into P_1 and P_2 defined as follows:

$$P_1 := \bigcup_{\chi \in X_2} FR(\chi, S) \quad \text{and} \quad P_2 := \bigcup_{\chi \in X_1} FR(\chi, S).$$

Then, we let $S'_1 := S_1 \cap P_2$ and $S'_2 := S_2 \cap P_1$. If $S'_1 = S'_2 = \emptyset$, then $S_1 \subset P_1$ and $S_2 \subset P_2$, so the lemma is true. Thus, in the following, we assume that $S'_1 \cup S'_2 \neq \emptyset$. Note that $S'_1 \cap X_1 = \emptyset$ and $S'_2 \cap X_2 = \emptyset$ by our choice of (S_1, S_2) .

Let $S_1^* := (S_1 \cup S'_2) \setminus S'_1$ and $S_2^* := (S_2 \cup S'_1) \setminus S'_2$. Then, (S_1^*, S_2^*) forms a bipartition with $S_1^* = S \cap P_1$ and $S_2^* = S \cap P_2$. We claim that (S_1^*, S_2^*) is indeed optimal. First, observe that for any $q \in S'_1$, q lies in $FR(\chi, S)$ for some $\chi \in X_1$, so we have $\Phi_{S_1}(q) = d(q, \chi)$ for some $\chi \in X_1$. On the other hand, we have $d(q, q') \leq d(q, \chi)$ for any $q' \in S$. This implies that $\Phi_{S_1}(q) \geq \Phi_{S_2^*}(q)$ for any $q \in S'_1$. Analogously, we have $\Phi_{S_2}(q) \geq \Phi_{S_1^*}(q)$ for any $q \in S'_2$.

Now, we consider $\text{diam}(S_1^*)$ and $\text{diam}(S_2^*)$. Since $\text{diam}(S_1^*) = \max_{q, q' \in S_1^*} d(q, q')$, we have $\text{diam}(S_1^*) = \max\{\text{diam}(S_1 \setminus S'_1), \max_{q \in S'_2} \max_{q' \in S_1^*} d(q, q')\}$. Then, since $\max_{q \in S'_2, q' \in S_1^*} d(q, q') = \max_{q \in S'_2} \Phi_{S_1^*}(q)$ and $\Phi_{S_2}(q) \geq \Phi_{S_1^*}(q)$ for any $q \in S'_2$, we have

$$\text{diam}(S_1^*) = \max \left\{ \text{diam}(S_1 \setminus S'_1), \max_{q \in S'_2} \Phi_{S_1^*}(q) \right\} \leq \max \left\{ \text{diam}(S_1 \setminus S'_1), \max_{q \in S'_2} \Phi_{S_2}(q) \right\}.$$

This implies that $\text{diam}(S_1^*) \leq \max\{\text{diam}(S_1), \text{diam}(S_2)\} = \text{diam}_2(S)$ since $S_1 \setminus S'_1 \subseteq S_1$ and $S'_2 \subseteq S_2$. Analogously, we also obtain the same bound for $\text{diam}(S_2^*)$: $\text{diam}(S_2^*) \leq \max\{\text{diam}(S_1), \text{diam}(S_2)\} = \text{diam}_2(S)$. Hence, we conclude that $\max\{\text{diam}(S_1^*), \text{diam}(S_2^*)\} = \text{diam}_2(S)$, and (S_1^*, S_2^*) is another optimal bipartition of S . $\square$

Our algorithm that computes a two-center of S is described as follows: First, compute the set X of extreme points of S , and the farthest-neighbor Voronoi diagram $\text{FVD}(S)$. For each bipartition (X_1, X_2) of X that satisfies the property of Lemma 8.2, let $S_1 := S \cap \bigcup_{\chi \in X_2} FR(\chi, S)$ and $S_2 := S \cap \bigcup_{\chi \in X_1} FR(\chi, S)$. Then, compute $\text{diam}(S_1)$ and $\text{diam}(S_2)$, and keep the minimum of $\max\{\text{diam}(S_1), \text{diam}(S_2)\}$ for all such bipartitions of X .

Let (S_1^*, S_2^*) be the bipartition of S with a minimum value of $\max\{\text{diam}(S_1^*), \text{diam}(S_2^*)\}$. Then, (S_1^*, S_2^*) is an optimal bipartition and $\text{diam}_2(S) = \max\{\text{diam}(S_1^*), \text{diam}(S_2^*)\}$ by Lemmas 8.1 and 8.3. A two-center (c_1, c_2) of S can be found by choosing any $c_1 \in \text{cen}(S_1^*)$ and any $c_2 \in \text{cen}(S_2^*)$.

The above algorithm works properly when S is a finite set of points in P .

Theorem 8.4 *Let S be a set of m points in a simple n -gon P . Then, an L_1 geodesic two-center of S can be computed in $O(n + m \log n)$ time.*

Proof The correctness follows from Lemma 8.3.

To check the time complexity, we apply Theorem 7.5 to compute X and $\text{FVD}(S)$. For each bipartition (X_1, X_2) of X , we find the corresponding bipartition (S_1, S_2) of S , and then compute $\text{diam}(S_1)$, $\text{diam}(S_2)$, $\text{cen}(S_1)$, and $\text{cen}(S_2)$ in $O(n + m \log n)$ time again by Theorem 7.5. Since $|X| \leq 4$ by Theorem 4.1, there are a constant number of different bipartitions (X_1, X_2) of X . Thus, all the effort is bounded by $O(n + m \log n)$ time. $\square$

Another interesting special case is when $S = P$.

Theorem 8.5 *An L_1 geodesic two-center of a simple n -gon can be computed in $O(n)$ time.*

Proof First, compute the set X of extreme points of P and $\text{FVD}(P)$ by applying Theorem 7.6. For each bipartition (X_1, X_2) of X , we construct

$$P_1 := \bigcup_{\chi \in X_2} FR(\chi, P) \text{ and } P_2 := \bigcup_{\chi \in X_1} FR(\chi, P).$$

This can be done in $O(n)$ time by traversing ∂P by Theorem 6.3. Then, observe that P_1 and P_2 are (weakly) simple polygons with a total of $O(n)$ vertices. Thus, $\text{diam}(P_i)$ and $\text{cen}(P_i)$ can be computed in $O(n)$ time for $i = 1, 2$ again by Theorem 7.6. Since $|X| \leq 4$ by Theorem 4.1, there are a constant number of different bipartitions (X_1, X_2) of X . Thus, all the effort is bounded by $O(n)$ time. $\square$

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